



Indonesian National Standard

SNI ISO 37122:2019
(Determined by BSN in 2019)

(ISO 37122:2019, IDT)

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Published in Jakarta

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Foreword

Indonesian National Standard (SNI) ISO 37122:2019, *Sustainable Cities and Communities – Indicators for Smart Cities*. This standard is a new standard that adopts ISO 37122:2019 *Sustainable Cities and Communities — Indicators for Smart Cities* with a monolingual translation method. This standard was developed to establish a methodology for a set of indicators for smart cities.

This SNI was prepared by Technical Committee 13-11, Sustainable Urban and Community, which held a technical meeting and reached a consensus on October 16, 2019 in Jakarta, attended by representatives from producers, consumers, testing institutions, universities, and other related agencies.

This standard has gone through a polling process from October 25, 2019 to November 24, 2019 with the results approved. Finally,

For your information, the term "city" in this standard refers to a city/district. If users have any doubts about this standard, they are advised to consult the original standard, ISO 37122:2019, and/or other related accompanying documents.

It should be noted that some elements of this standard document may be patented. The National Standardization Agency is not responsible for identifying any or all existing patents.

Foreword

ISO (International Organization for Standardization) is a federation (union) of national standards bodies throughout the world (ISO member bodies).

The work of developing International Standards is usually carried out by ISO technical committees. Any member body interested in a subject for which a technical committee has been established has the right to be represented on that committee.

International organizations, both governmental and non-governmental, in conjunction with ISO, also participate in this work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters relating to electrotechnical standardization.

The procedures used to develop this document and the procedures intended for its further maintenance are described in ISO/IEC Directives, Part 1. It is particularly important to be aware of the different approval criteria required for the various types of ISO documents. This document has been prepared in accordance with the editorial rules of ISO/IEC Directives, Part 2 (see www.iso.org/directives).

It should be noted that some elements of this document may be the subject of patent rights. ISO is not responsible for identifying any or all such patent rights. Details of any patent rights identified during the development of this document will be set out in the Introduction and/or in ISO's list of patent claims received (see www.iso.org/patents).

Any trade names used in this document are for informational purposes only and are provided for the user's convenience and do not constitute an endorsement.

For an explanation of the meaning of ISO-specific terms and expressions related to conformity assessment, and also for information on ISO's compliance with the WTO principles on Technical Barriers to Trade (TBT), please visit the following URL: www.iso.org/iso/foreword.html.

Any feedback or questions on this document should be directed to the user's national standards body. A complete list of these bodies can be found at www.iso.org/members.html.

Introduction

- Apply collaborative leadership methods, working through discipline and systems city;
- using data information and modern technology to provide better services and quality of life to those in the city (residents, businesses, visitors);
- Provide a better living environment where smart policies, practices and technologies are used to serve citizens; – achieve their sustainability and environmental goals in a more innovative;
- identifying the needs and benefits of smart infrastructure; – facilitating innovation and growth; – building a dynamic and innovative economy that is ready to face challenges future.

Sustainable cities and communities – Indicators for smart cities

1. Scope

This document defines and establishes definitions and methodologies for a set of indicators for smart cities.

Accelerating improvements in services and quality of life is a key part of the definition of a smart city, this document, in conjunction with ISO 37120, is intended to provide a comprehensive set of indicators to measure progress towards a smart city as represented by Figure 1.

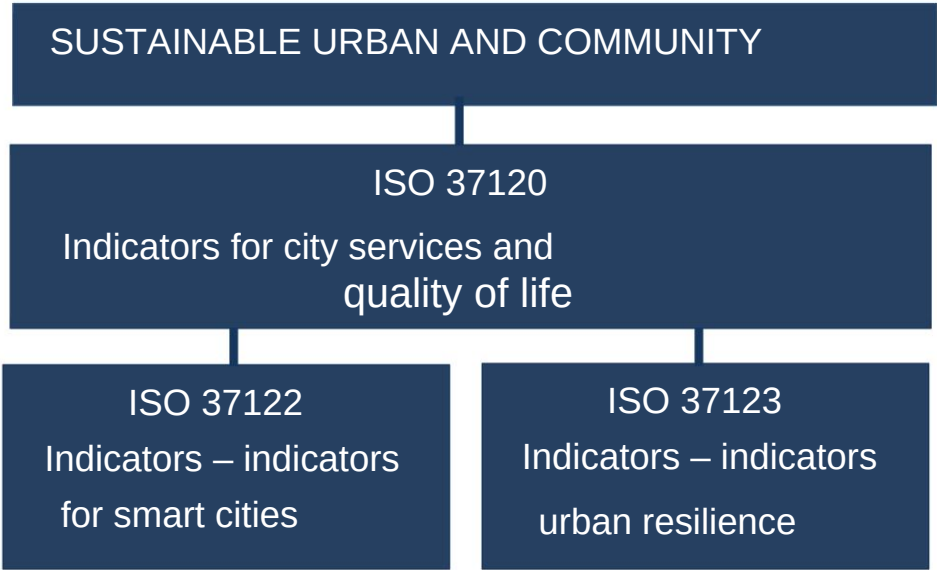


Figure 1 - Sustainable community development - Relationship between the standard family of smart city indicators

2. Normative References

The following documents are referred to in the text in such a way that some or all of their content may be used in conjunction with this document. For dated references, only the cited edition applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

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3. Terms and Definitions

For the purposes of this document, the terms and definitions given in ISO 37101, ISO 37120 and the following apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses: - ISO online search platform :

available at <https://www.iso.org/obp> - IEC Electropedia: available at <http://www.electropedia.org/>

3.1.

gigajoule a

measure of energy equal to 1 billion Joules (J), where 1 J is the amount of energy required to send an electric current of one ampere through a resistance of one ohm for one second.

NOTE 1 to entry: One gigajoule (GJ) is equal to 277.8 kilowatt hours (kWh)

3.2.

per 100,000 population for
every 100,000 city residents

NOTE 1 to entry: The choice of 100,000 inhabitants was made to allow cities of different sizes to compare results with each other relatively easily and effectively. It should be noted that in some countries statistics are collected per 1,000 capita and slight mathematical adjustments may be necessary to reflect this difference for accurate comparisons. The per 1,000 population measure may be a more applicable measure for smaller cities.

3.3.

public buildings State-

owned or leased buildings that function as government administrative offices, libraries, recreational facilities, hospitals, schools, fire stations or police stations.

NOTE 1 to entry: Building ownership (public or private) is defined differently according to region and political system. The restricted definition used here allows for global comparability across cities.

3.4.

Smart city a

city that increases its pace in delivering social, economic, and environmental sustainability outcomes and responds to challenges such as climate change, rapid population growth, and political and economic instability by fundamentally improving how it engages with the community, applying collaborative leadership methods, working through disciplines and city systems, and using data information and modern technologies to provide better services and quality of life to the people of the city (residents, businesses, visitors), now and for the future, without unfairly harming others or degrading the natural environment.

NOTE 1 to entry: smart cities also face the challenge of respecting planetary boundaries and taking into account the limitations these boundaries impose.

NOTE 2 to entry: There are many definitions for smart city; however, the definition used in TC268 is the official definition approved by the ISO *Technical Management Board*.

3.5.

Biosolids

are a mixture of water and solids separated from various types of water as a result of natural or artificial processes.

4. Urban indicators

This document is designed to assist cities in guiding and assessing the performance management of urban services and all service provisions, as well as quality of life. It considers sustainability as a general principle, and intelligence and resilience as guiding concepts in urban development. All indicators must be compiled annually. If census data is used, the census year must be reported. The year of population data must also be reported.

For data interpretation purposes, cities should consider contextual analysis when interpreting the results. The local institutional environment can influence the capacity to implement indicators. In some cases, services may be provided by the private sector or the community itself.

The list of indicators is based on the following criteria:

- Completeness: indicators must be able to measure and balance all aspects relevant for smart city evaluation.
- Technology neutral: not biased towards any technology, whether existing now or in the future will come.
- Simplicity: indicators can be expressed and presented in a clear and understandable manner.
- Validity: indicators are an accurate reflection of facts and data that can be collected using scientific techniques.
- Verifiable: the indicator can be verified and reproduced. The methodology is quite rigorous to provide certainty at the level of implementation of the criteria.
- Availability: the quality of the data is available, or is feasible to initiate a secure and reliable monitoring process that will make it available in the future.

When interpreting results from a specific service area, it is important to review results from a variety of indicators across themes; focusing on a single indicator can lead to distorted or incomplete conclusions. Aspirational elements should also be considered in the analysis.

Users may also consider the following aspects that should be clearly stated in the report and justified: indicators can be aggregated to larger administrative areas (e.g., regions, metropolitan areas); indicators can be grouped for analysis when considering the holistic characteristics of the city; and, this set of indicators can be complemented with other sets of indicators to have a more comprehensive holistic approach to smart and sustainable city analysis.

In addition, it is also important to be aware of the potential conflicting impacts of certain indicator results, whether positive or negative, when analyzing the results.

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Data sources may vary depending on the city and may differ from those mentioned in this document. However, data must be verifiable, auditable, trustworthy, and justified. Cities may not have access to all the data required for the indicators in this document because services are provided by third parties. However, it is still important for cities to obtain this data. A crucial component of smart cities is the role of public/private partnerships, and this collaboration, including data sharing, must be supported.

Cities using this document must report at least 50% of the indicators in this document as it is intended to be used in conjunction with ISO 37120. "Smart cities" is a relatively new and evolving concept aimed at cities worldwide and it is important for cities to report progressively more indicators in this document over time.

5. Economy**5.1. Percentage of service contracts providing municipal services that include open data policy****5.1.1. General**

Those implementing this document must report these indicators in accordance with the following requirements.

NOTE 1 An open data policy demonstrates a city's commitment to better managing business information throughout the information lifecycle. Identifying and making data accessible helps ensure that the public is informed and engaged through transparent, accountable, and accessible government.

NOTE 2 This indicator reflects the issues of "Governance, empowerment and involvement", "Innovation, Creativity and research" and "Economy and sustainable production and consumption" as defined in ISO 37101. This can allow the evaluation of the contribution to the "Attractiveness" and "Responsible use of resources" objectives of the city as defined in ISO 37101.

5.1.2. Indicator requirements

The percentage of service contracts providing city services that contain an open data policy shall be calculated by dividing the total number of service contracts providing city services that contain an open data policy (the numerator) by the total number of service contracts in the city (the denominator). The result shall then be multiplied by 100 and expressed as a percentage of service contracts providing city services that contain an open data policy.

Open data policy refers to data that will be provided and used by the city where the data can be analyzed and is open to the public, including residents and non-residents of the city.

Service contracts refer to agreements with businesses that provide city services.

City services should refer to services provided by the city and typically include the following areas: general services, trash and recycling; public safety; fire; streets and traffic; recreation; construction; regulations, violations and enforcement; permits and licenses; planning; buildings; policies, projects and initiatives; city building rentals and services; water, wastewater and sewer; and property taxes and utilities.

5.1.3. Data Sources

Data derived from service contracts should be sourced from relevant urban agencies, or market research and survey companies.

5.1.4. Data Interpretation

A higher percentage of service contracts providing city services, with open data availability, leads to greater transparency in the performance of city services and technologically advanced communities. This allows people to review the data and performance of businesses contracted by the city to provide city services that are typically not covered by the city's published performance measures.

5.2. New business continuity rate per 100,000 population

5.2.1. General

NOTE 1 New businesses make a positive contribution to the local economy, and start-up activity can signal a city's economic potential. New businesses have the potential to contribute a significant number of new jobs to the economy and tend to have faster employment growth rates, particularly in innovation-driven/technology-focused companies, such as computer or software development.

NOTE 2 This indicator reflects the issues of "Economy and sustainable production and consumption" and "Innovation, creativity and research" as defined in ISO 37101. This can allow the evaluation of the contribution to the "Attractiveness" and "Resilience" objectives of cities as defined in ISO 37101.

5.2.2 Indicator requirements

The new business survival rate per 100,000 population should be calculated by dividing the number of new businesses in the city (the numerator) by 1/100,000 of the city's total population (the denominator). The result should be expressed as a new business survival rate per 100,000 population.

The survival rate refers to new businesses that have been created during the last two years, registered in the city and still operating during the last year (numerator) divided by the total number of new businesses that have been created during the last two years and registered in the city (denominator).

Business should refer to a company or firm located in the city.

These businesses may be considered to have innovative business processes and/or products. Cities reporting this indicator must specify the sectors and categories of innovative businesses included in this calculation. A company is the smallest combination of legal units, which

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An innovative business is an organizational unit that produces goods or services. Businesses can be categorized as simple (one operating entity) or complex (multiple operating entities) as stated in ISO 37120:2018, 5.5.2. An innovative business refers to a business that implements new ideas, creates dynamic products, or improves existing services in any industry.

5.2.3. Data sources

New business data must be obtained through the relevant urban agency or ministry that oversees the approval of new business licenses or registration of new businesses.

5.3. Percentage of workforce working in the Information Technology Sector and Communication (ICT)

5.3.1 General

Those implementing this document must report on these indicators in accordance with the following requirements.

NOTE 1 The information and communications technology (ICT) sector is a combination of manufacturing and service industries that capture, transmit and display data and information electronically[8]. With the rapid development of ICT and the mainstreaming of ICT into everyday life, the link between ICT technology and human resource development is very clear. ICT has long been recognized as a key driver for bridging the digital divide and achieving the three dimensions of sustainable development—economic growth, environmental balance, and social inclusion. and promote innovation in society[7]. Thus, having a workforce to develop the ICT sector will help drive economic growth, increase labor productivity and enhance international competitiveness through innovative ICT development, including through wider use of ICT products and services across sectors.

economy and society.

NOTE 2 This indicator reflects “Innovation, creativity and research”, “Education and capacity building” and “Economy and sustainable production and consumption” as defined in ISO 37101. This can enable the evaluation of the contribution to the “Attractiveness” and “Resilience” objectives of cities as defined in ISO 37101.

5.3.2 Indicator requirements

The percentage of the workforce employed in ICT sector jobs is calculated by dividing the number of city residents employed in the ICT sector (numerator) by the city's total workforce (denominator). The result is then multiplied by 100 and expressed as a percentage of the labor force employed in ICT sector jobs.

The ICT sector refers to a combination of manufacturing and service industries that capture, transmit, and display data and information electronically. For manufacturing industries, products from candidate industries must be intended to fulfill information processing and communication functions, including transmission and display, and must use electronic processing to detect, measure, and/or record physical phenomena or control physical processes. For service industries, products from candidate industries must be intended to enable information processing and communication functions by electronic means.

More specifically, the ICT sector refers to *the United Nations Statistics Division's International Standard Industrial Classification of All Economic Activities (ISIC) Rev. 4* (https://unstats.un.org/unsd/publication/seriesm/seriesm_4rev4e.pdf) section J (Information and communication) as a whole, and subsections of section C (Manufacturing) which

in line with the ICT manufacturing industry mentioned above, such as sections 26 (Computer, electronic and optical products industry) and 27 (Electrical equipment industry).

Labor force should refer to, as defined by *the International Labour Organization (ILO)*, the total labor force consisting of employed and unemployed workers. Therefore, working age should refer to all persons equal to or older than the legal working age in the reference jurisdiction. This indicator should exclude child labor, which is defined as workers aged 14 years or younger.

5.3.3 Data sources

Data on employment by industry must be obtained through labor force surveys or municipal employment assessments administered by local, regional or national statistical authorities/agencies, or the Ministry or Department of Manpower.

5.4. Percentage of the workforce working in the education, research and development sector

5.4.1 General

Those implementing this document must report these indicators in accordance with the following requirements.

NOTE 1 As cities and communities strengthen their focus on developing their knowledge economy, the role of education and research and development industries is paramount in human resource development. These industries play a vital role in economic development, promoting innovative thinking processes to improve existing products or services or to develop new ones.

In addition, the education sector includes those employed at all levels of the education system, ensuring that citizens have access to education and receive effective services.

Thus, the workforce employed in these two industries helps in the development or improvement of products and services, in addition to ensuring that citizens receive high-quality education to actively engage in the knowledge economy.

NOTE 2 This indicator reflects "Innovation, creativity and research" and "Economy and sustainable production and consumption" as defined in ISO 37101. This can allow the evaluation of the contribution to the "Attractiveness" and "Resilience" objectives of cities as defined in ISO 37101.

5.4.2 Indicator requirements

The percentage of the workforce employed in the education, research, and development sector is calculated by dividing the number of city residents employed in the education, research, and development sector (the numerator) by the city's total workforce (the denominator). The result is then multiplied by 100 and expressed as a percentage of the workforce employed in the education, research, and development sector.

When defining education and research and urban development sectors will refer to *the United Nations Statistics Division's International Standard Industrial Classification of All Economic Activities (ISIC) Rev. 4* (https://unstats.un.org/unsd/publication/seriesm/seriesm_4rev4e.pdf).

The education sector refers to organizations that are primarily engaged in providing instruction and training in various subjects. Instruction and training are provided by

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specialized institutions, such as schools, colleges, universities and training centers. The City will refer to any organization included in section P of ISIC, Rev.4, which is divided into the following subsections: 851 (Pre-primary and primary), 852 (Secondary education), 853 (Higher education), 854 (Other education) and 855 (Education support activities).

The research and development section refers to organizations primarily engaged in conducting preliminary investigations, conducted systematically to gain new knowledge (research), and in applying research findings or other scientific knowledge to the creation of new or significantly improved products or processes (experimental development). The City will refer to organizations included in Division 72 - Scientific research and development section M - Professional, scientific and technical activities of ISIC, Rev.4.

The numerator of this indicator should be calculated from the sum of the total number of people employed in these two sectors, education and research and development.

Labor force should refer to the total labor force, as defined by the International Labour Organization (ILO). Therefore, working age should refer to all persons equal to or older than the legal working age in the reference jurisdiction. This indicator should exclude child labor, which is defined as workers aged 14 years or younger.

5.4.3 Data sources

Data on employment by industry must be obtained through labor force surveys or municipal employment assessments administered by local, regional or national statistical authorities/agencies, or the Ministry or Department of Manpower.

6. Education**6.1. Percentage of the city's population with professional skills in more than one Language****6.1.1 General**

Those implementing this document must report these indicators in accordance with the following requirements.

NOTE 1 Foreign language skills are one of the distinguishing features of a work-ready workforce. This must also be supported by a successful educational program.

NOTE 2 This indicator reflects the issue of "Education and capacity development" as defined in ISO 37101. It can enable the evaluation of the contribution to the "Social cohesion", "Well-being", "Attractiveness" and "Resilience" objectives of the city as defined in ISO 37101.

6.1.2 Indicator requirements

The percentage of a city's population with professional proficiency in more than one language is calculated of the total number of people who can communicate in more than one foreign language with professional proficiency (numerator) divided by the total population of the city (denominator). The result will then be multiplied by 100 and expressed as the percentage of the city's population with professional proficiency in more than one language.

A foreign language should refer to a language spoken professionally other than one of the official languages of the country in which the city is located. For example, the official language of the United States (US) is English, so city residents who speak a language other than English professionally are counted as speaking a foreign language in the United States. In the case of countries with more than one official language, such as Canada with two designated official languages (English and French), when a person can speak both official languages at a professional proficiency level, one official language should be included in the foreign language count and the other excluded from the foreign language count. For example, if someone in Canada speaks both English and French, they would have one foreign language and it would be counted in the numerator of this indicator. Similarly, if a Canadian speaks only English (one of the official languages) but also has professional skills in Spanish, for example, then that person would be considered to have professional proficiency in more than one language.

Professional proficiency refers to the following levels of competency:

- Able to speak the language with sufficient structural accuracy, vocabulary and structural coherence in discourse to participate effectively in most formal and informal conversations on practical, social and professional topics; understanding on a complete basis; Can discuss fluently and easily abstract problems and areas of
- competence and special interest; can support
- opinions and hypothesize; can provide clear and well-organized structured arguments; while the influence of the speaker's first language may be
- noticeable (in pronunciation, grammar and vocabulary),
- there should be no patterned errors, and errors should not distract the listener or interfere with
- communication.

For reference, the above definition of professional proficiency corresponds to level C1 of *the Common European Framework of Reference for Languages: Learning, Teaching, Assessment*.

6.1.3 Data sources

Data on foreign languages spoken by a city's population should be sourced using census data, or local, regional or national surveys relating to the languages spoken.

6.1.4 Data interpretation

A high percentage of residents who can communicate in more than one foreign language indicates that the city has an educated and diverse population that can handle interactions that transcend national boundaries. Globalization, economic growth in developing countries, and improvements in transportation infrastructure have resulted in major changes in world trade patterns, and a higher proportion of the world's population is able to visit other countries, whether for business, leisure, or other reasons.

Foreign language skills have the potential to enhance one's mobility, employability and personal development.

6.2. Number of computers, laptops, tablets or other digital learning devices available per 1,000 students

SNI ISO 37122:2019**6.2.1 General**

Those implementing this document must report these indicators in accordance with the following requirements.

NOTE 1 Computer literacy is an important aspect of professional employability in many sectors, and enables alternative forms of civic engagement. Increasing the accessibility of electronic devices for students, as well as exposure to computers, laptops, tablets or other digital learning devices, can improve students' computer literacy. This also enables citizens to access a wider range of information, empowering people from all walks of life to search, evaluate, use and create information effectively to achieve personal, social, occupational and educational goals.

NOTE 2 This indicator reflects the issue of "Education and capacity development" as defined in ISO 37101. It can allow the evaluation of the contribution to the objective of "Attractiveness" of the city as defined in ISO 37101.

6.2.2 Indicator Requirements

The number of computers, laptops, tablets, or other digital learning devices available per 1,000 students shall be calculated as the total number of computers, laptops, tablets, or other digital learning devices with Internet access available to elementary and secondary school students attending elementary and secondary schools in the city (numerator) divided by 1/1,000 of the total elementary and secondary school population (denominator). The result shall be expressed as the number of computers, laptops, tablets, or other digital learning devices available per 1,000 students.

Only schools that own/provide computers, laptops, tablets or other digital learning devices are counted.

Primary school students shall refer to students enrolled in primary education as defined in ISO 37120:2018, Clause 3.

Secondary school students shall refer to students enrolled in secondary education as defined in ISO 37120:2018, Clause 3.

The number of computers, laptops, tablets or other digital learning devices available per 1,000 students should be reported separately for primary and secondary school students in the table below.

	Number of computers, laptops, tablets or other digital learning devices	Number of computers, laptops, tablets or other digital learning devices per 1,000 students
Elementary School Students		
Student School		
Advanced		
Total Students (School Elementary and School Advanced)		

6.2.3 Data sources

Data on the number of electronic devices with internet access should be sourced from local school leaders, or the Ministry or Department of Education.

6.2.4 Data interpretation

While computer literature can be beneficial for secondary school students, there is debate about the positive impact of the availability of digital devices and technology for primary school students.

Sufficiently fast broadband, as defined in 18.1, may also be considered for analysis of this indicator. When a low percentage of a city's population has fast broadband, the benefits of accessibility to computers, laptops, tablets, or other digital learning devices may be limited. Sufficiently fast broadband refers to a network with a speed of at least 256 kbit/s in both directions, uploading and downloading. This speed is sufficient for internet browsing and email. Sufficient bandwidth corresponds to basic broadband.

6.3. Number of science, technology, engineering and mathematics (STEM) higher education levels per 100,000 population

6.3.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1: Receiving a higher education provides individuals with a foundation for meaningful participation in the workforce and helps reduce poverty and inequality. This pillar of human development is widely recognized as a key pathway to social mobility. All disciplines taught by higher education institutions benefit society in some way, such as science, technology, engineering, and mathematics (STEM) disciplines, which are crucial for the technological development and innovation of cities. STEM education helps create critical thinkers, improves scientific literacy, and enables the next generation of innovators. Furthermore, STEM is important because science permeates every part of our lives, and the need for STEM degree holders is increasing with the growing demand for product and process innovators who will help sustain and promote economic growth.

NOTE 2 This indicator reflects the "Education and capacity development" and "Innovation, creativity and research" issues as defined in ISO 37101. This can enable the evaluation of the contribution to the "Resilience" and "Attractiveness" objectives of cities as defined in ISO 37101.

6.3.2 Indicator requirements

The number of STEM higher education degrees per 100,000 population should be calculated as the number of people holding a higher education degree with a specialization or discipline major in a STEM subject (numerator) divided by 1/100,000 of the total city population (denominator).

The results will be expressed as the number of STEM higher education degrees per 100,000 population.

STEM degrees refer to higher education degrees specializing in subjects within the fields of science, technology, engineering, and mathematics, and are intended to address a wide range of educational and employment opportunities, beyond the narrower fields of science and mathematics. STEM programs are typically classified into several occupational groups: computer science and technology; mathematical sciences; digital music and digital arts; engineering and surveying; and natural, physical, and life sciences.

This indicator will only cover city residents, and will not include temporary residents or international students.

Higher education refers to the definition of higher education stated in ISO 37120:2018, Clause 3.

6.3.3 Data sources

Data on higher education levels by subject must be sourced from local universities/institutions issuing diploma certificates, or the Ministry or urban institutions related to education, if available. If higher education data from these sources is not available, data from surveys or censuses can be used.

6.3.4 Data interpretation

This indicator provides an overview of the population's skill set. This data can also impact surrounding cities, as individuals with STEM degrees can work in those cities or, in other cities, contexts can force less-educated individuals to move to surrounding cities, creating intellectual *ghettos* within the city. While only STEM disciplines are considered for this indicator, social science and other disciplines are also important to a city's workforce and can contribute to its intelligence.

7. Energy

7.1. Percentage of electrical energy and thermal energy generated from wastewater treatment, solid waste and other liquid waste treatment and other waste heat resources, as a share of the city's total energy mix for a given year

7.1.1 General

Those who apply this document must report these indicators in accordance with the following requirements.

NOTE 1 Waste heat is an endogenous energy source in every city. Waste heat can be obtained from wastewater and solid waste treatment plants or other industrial processes, as well as from the tertiary sector and transportation (e.g., heat rejected from data centers or subway ventilation).

Wastewater is a renewable resource that carries heat and chemical energy. In some cases, wastewater has been found to contain nearly five times the energy required to process and treat it. It is crucial for cities to recognize wastewater's potential as a sustainable energy source and utilize it in their energy mix.

Additionally, wastewater treatment plants consume significant amounts of energy and produce greenhouse gas emissions. Wastewater treatment plants also have the potential to be a renewable energy source for cities. Wastewater treatment plants can use wastewater to generate on-site energy, which can be used to help operate the wastewater treatment plant, thereby reducing wastewater treatment plant operating costs, energy consumption, and greenhouse gas emissions.

While waste reduction, recycling, and composting can reduce the environmental impact of municipal solid waste, not all types of materials can be practically and economically recycled in an environmentally beneficial manner.

These residual solids can provide opportunities to generate energy, using new and cleaner technologies.

Waste heat can be categorized as high-enthalpy and low-enthalpy. While high-enthalpy waste heat enables electricity production, low-enthalpy waste heat can be used directly to heat (and/or cool) buildings, typically through area-located heating and cooling networks.

NOTE 2 This indicator reflects the issues of “Economy, sustainable production-consumption” and “societal infrastructure” as defined in ISO 37101.

This can enable evaluation of the contribution of the city to the “Responsible use of resources” objective as defined in ISO 37101.

7.1.2 Indicator Requirements

The percentage of electrical and thermal energy generated from wastewater treatment, solid waste and other liquid waste treatment and other waste heat resources as part of the total energy mix of the city for a given year, shall be calculated from the total amount of electrical and thermal energy expressed in GJ (*Gigajoules*) generated from wastewater treatment, solid waste and other liquid waste treatment and other waste heat resources (numerator) divided by the total energy demand of the city's end users in the same units as the numerator (GJ). The result is then multiplied by 100 and expressed as the percentage of electrical and thermal energy generated from wastewater treatment, solid waste and other liquid waste treatment and other waste heat resources as part of the total energy mix of the city for a given year.

The term energy mix refers to the combination of different primary energy sources used to meet the energy needs of a particular geographic area.

Where possible, data from wastewater treatment, solid waste and other liquid waste treatment plants, and other waste heat sources should be included and listed separately. This should include the use of electrical and thermal energy generated on-site at wastewater/wastewater facilities and off-site use by external parties.

	Thermal Energy (% Mixed Energy Total)	Electrical Energy (% Mixed Energy Total)	Total Energy (% Mixed Energy Total)
Wastewater Sources			
Sources of Solid Waste			
Waste Heat Source			
Other			
Sources of Liquid Waste			
Other			
Total (%)			

Waste heat should be considered as all the remaining heat energy generated in the city, which is not used, as well as potential chemical energy sources that are not valued as energy.

Wastewater treatment refers to the physical, chemical, and biological processes used to remove, reduce, or neutralize contaminants from wastewater before it is discharged into water bodies. Wastewater treatment can include primary, secondary, or tertiary treatment, or even higher standards of wastewater treatment.

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Solid waste treatment shall refer to the physical, chemical, and biological processes used to remove, reduce, or neutralize contaminants from solid waste before recycling, recovery, or disposal.

Other liquid waste refers to liquid waste such as fat, oil or grease *which* are energy sources.

7.1.3 Data sources

Data on the amount of electrical and thermal energy generated from wastewater treatment, solid waste and other liquid waste treatment and other waste heat resources must be sourced from the relevant regional government organizations or ministries that oversee these matters, as well as from local regulators and utility service providers.

7.2. Electrical and thermal energy (GJ) generated from wastewater treatment per capita per year**7.2.1 general**

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Wastewater is a renewable resource that carries both heat and chemical energy. In some cases, wastewater has been found to contain nearly five times the amount of energy required to process and treat the wastewater. It is important for cities to recognize the potential of wastewater as a sustainable energy source and utilize it in their energy mix.

While wastewater treatment plants use significant amounts of energy and produce greenhouse gas emissions, they also have the potential to be a renewable energy source for cities. Wastewater treatment plants can use wastewater to generate on-site energy that can be used to operate the wastewater treatment plant, thereby reducing wastewater treatment plant operating costs, energy consumption, and greenhouse gas emissions.

NOTE 2 This indicator reflects the issues of "Economy, sustainable production-consumption" and "community infrastructure" as defined in ISO 37101. This can allow an evaluation of the contribution of the city to the objective "Responsible use of resources" as defined in ISO 37101.

7.2.2 Indicator Requirements

The electrical and thermal energy (GJ) generated from wastewater treatment per capita per year shall be calculated by dividing the total amount of electrical and thermal energy expressed in GJ generated from wastewater treatment in the city (numerator) by the total population of the city (denominator). The result shall be expressed as the sum of electrical and thermal energy in GJ generated from wastewater treatment per capita for a given year.

This should include the use of electrical and thermal energy generated on-site at wastewater facilities and off-site use by external parties.

Wastewater treatment refers to the physical, chemical, and biological processes used to remove, reduce, or neutralize contaminants from wastewater before it is discharged into water bodies. Wastewater treatment can include primary, secondary, or tertiary wastewater treatment, or wastewater treatment to higher standards.

7.2.3 Data Sources

Data on the amount of electrical and thermal energy generated from wastewater treatment, solid waste and other liquid waste treatment and other waste heat resources must be sourced from the relevant regional government organizations or ministries that oversee these matters, as well as from local regulators and utility service providers.

7.3. Electrical and thermal energy (GJ) is generated from solid waste or waste processing. liquid per capita per year

7.3.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 While waste reduction, recycling and composting can reduce the environmental impact of municipal solid waste, not all types of materials can be practically and economically recycled in an environmentally beneficial manner. Therefore, these residual solid wastes can provide opportunities for energy generation, using new and cleaner technologies. Other liquid wastes such as fats, oils and grease are also sources of energy.

NOTE 2 This indicator reflects the issues of "Economy, sustainable production-consumption" and "community infrastructure" as defined in ISO 37101. This can allow an evaluation of the contribution of the city to the objective "Responsible use of resources" as defined in ISO 37101.

7.3.2 Indicator Requirements

The electrical and thermal energy (GJ) generated from solid waste or liquid waste processing per capita per year shall be calculated as the total amount of electrical and thermal energy expressed in GJ generated from solid waste and other liquid waste processing in the city (numerator) divided by the total population of the city (denominator). The result shall be expressed as the amount of electrical and thermal energy in GJ per capita for a given year.

This should include the use of electrical and thermal energy generated on-site at the solid waste facility and off-site use by external parties.

This also includes other liquid waste such as fat, oil or grease .

7.3.3 Data Sources

Data on the amount of electrical and thermal energy generated from wastewater treatment, solid waste and other liquid waste treatment and other waste heat resources must be sourced from the relevant regional government organizations or ministries that oversee these matters, as well as from local regulators and utility service providers.

7.4. Percentage of city electricity produced using decentralized electricity production systems

7.4.1 General

Those who apply this document must report these indicators in accordance with the following requirements.

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NOTE 1 The primary benefit of a more decentralized energy system is the potential for a more reliable electricity system and higher-quality power supply (with appropriate support systems in place). A decentralized electricity production system can be defined as locating electricity production closer to the location of consumption, such as locating electricity production facilities within a city rather than sourcing electricity from regional electricity production facilities that are likely to be far from the city. While this approach is relatively new to the electricity industry and utilities, decentralized systems have the potential to lead to more optimal use of renewable energy sources, which in turn can reduce fossil fuel use and improve a region's energy efficiency and sustainability. Therefore, monitoring the amount of distributed electricity production can be used to assess a region's potential to utilize renewable energy sources and expand access to clean energy services, such as combined heat and power, that may be unavailable due to distance from central power generation facilities. The potential for utilizing heat to power is important—but this generally must be addressed during the initial design phase and is difficult to achieve otherwise.

NOTE 2 In addition to the environmental issues of decentralized electricity production mentioned previously, namely the adoption of renewable energy sources and the reduction of primary energy use, distributed electricity production can also be beneficial for the resilience of a city's electricity grid, especially during crisis events such as major storms.

NOTE 3 This indicator reflects the issues of “Economy, sustainable production-consumption” and “community infrastructure” as defined in ISO 37101. This can allow the evaluation of the contribution of the objectives “Responsible use of resources” and “Resilience” of cities as defined in ISO 37101.

7.4.2 Indicator Requirements

The percentage of urban electricity generated using decentralized electricity production systems is calculated by dividing the amount of electricity generated by decentralized electricity production systems/facilities in GJ (numerator) by the total amount of electricity consumed in the urban area in the same units as the numerator (GJ) - this includes electricity produced by both centralized and decentralized electricity production facilities (denominator). The result should be multiplied by 100 and expressed as a percentage of urban electricity, which is produced using a decentralized electricity production system.

Decentralized electricity, also known as distributed electricity generation, refers to the production of electricity at or near the point of use, regardless of size, technology, or fuel used—whether off-grid or on-grid. Furthermore, decentralized electricity generation encompasses a wide range of technologies, including wind turbines or photovoltaic (solar) panels, microturbines, and modular internal combustion engines.

7.4.3 Data Sources

Data on the amount of electricity generated by centralized and decentralized electricity production systems/facilities should be sourced from local utilities, and the relevant regional government agencies or ministries overseeing energy production. It is acceptable to use estimates based on energy audits conducted at sample sites to account for electricity generated immediately for automatic consumption, which is sometimes not communicated to the electricity service provider.

7.4.4 Data Interpretation

If one generator produces 100% of the city's electricity, this is not considered decentralized and the indicator is equal to 0%.

7.5. City energy network storage capacity per city's total energy consumption

7.5.1 General

Those who apply this document must report these indicators in accordance with the following requirements.

NOTE 1 Peak energy demand is a rarely discussed factor that increases energy costs and limits renewable energy penetration. Smart grids will accommodate energy storage (typically electrical and thermal storage, but also "clean" fuels such as hydrogen) to mitigate peak demand and shift energy use to intermittent periods of renewable energy production. Efficient storage capacity is crucial for balancing energy supply and demand in a region, and can be achieved through several strategies:

- electrical storage, on-site or nearby, including storage of chemicals such as hydrogen;
- electricity storage for electric vehicles (excluding hybrid vehicles); – thermal storage (e.g. hot and cold storage in heating and cooling schemes);
cooling of the nearest area locus);
- geothermal energy storage (e.g. some district heating and cooling schemes)
driven by a heat/cooling pump);
- the thermal mass of a building is considered thermal storage on site when associated with
predictive heat consumption algorithms that enable reduction of heat and cold peak demand from area locus heating and cooling schemes;
- other means of storing renewable energy such as fuel cells, where justified, including on-site energy storage.

NOTE 2 This indicator reflects the issues of "Economy, sustainable production-consumption" and "community infrastructure" as defined in ISO 37101. This can allow the evaluation of the contribution of the objectives "Responsible use of resources" and "Resilience" of cities as defined in ISO 37101.

7.5.2 Indicator Requirements

The storage capacity of the urban energy network per total urban energy consumption shall be calculated by dividing the total amount of energy that can be stored annually in the electricity network and urban thermal network (locus area heating and cooling schemes) in gigajoules (GJ) (numerator) by the total urban energy consumption (denominator). The result shall be expressed as the storage capacity of the urban energy network per total urban energy consumption.

Energy storage refers to the process of converting energy into a stored form that can later be converted back into energy when needed. Therefore, storage capacity refers to the amount of energy that can be stored.

For total end-user energy consumption, see ISO 37120:2018, 7.1.

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7.5.3 Data Sources

Data for storage capacity should be sourced from the relevant urban agency or ministry that oversees the urban energy network, and the local utility that operates the district heating and cooling scheme.

7.5.4 Data Interpretation

Having energy storage capacity near or within city limits (close to the load) reduces transmission losses and ensures a more reliable power supply. It also allows cities to better manage peak energy demand. If an energy storage facility is not located within city limits, the city must report the distance to the energy storage facility.

7.6. Percentage of street lighting managed by a light/lamp performance management system

7.6.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Remotely managed light points contribute to higher energy efficiency and can be optimized and customized to switch on and off and dim in each area of the city. Furthermore, remotely managed lights have the potential to improve safety in cities, so that if the light points facing the street are not sufficiently illuminated, this can be immediately monitored and localized to ensure rapid repairs. Finally, the actual energy consumption per light point can be accurately measured and reported with a light management system, to better monitor energy costs and CO2 reduction schemes.

NOTE 2 This indicator reflects the issues of "Economy, sustainable production-consumption" and "Safety and security" as defined in ISO 37101. This can allow an evaluation of the contribution of the city to the objective "Responsible use of resources" as defined in ISO 37101.

7.6.2 Indicator Requirements

The percentage of street lighting managed by a light performance management system shall be calculated by dividing the number of light points that can be controlled by the light performance management system (numerator) by the total number of light points in the city (denominator). The result shall then be multiplied by 100 and expressed as the percentage of street lighting managed by the lighting management system.

Controlled by a light performance management system should refer to the ability to monitor light points, set schedules for turning them on/off, and adjust light levels by dimming. This means that light points can be changed individually and remotely using an ICT-based system, which is connected via a communications network to the light points. The system should also be able to accurately measure the electrical energy consumed by the light points and indicate to the operator via the ICT-based system any failures that affect the light point's luminance.

7.6.3 Data sources

Data on remote lighting management should be sourced from regional government agencies or ministries responsible for street lighting inventory and street light management.

7.7. Percentage of street lighting that has been restored and newly installed

7.7.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Street lighting consumption can account for 15% to 50% of a city's total electricity consumption. Repairing city streetlights and installing new lighting can help improve energy efficiency, thereby reducing street lighting energy consumption. In addition, the recent market introduction of energy-efficient technologies for street lighting offers significant cost savings with a relatively short payback period. The savings in energy and annual maintenance costs may well cover the investment and capital costs.

NOTE 2 This indicator reflects the issues of "Economy, sustainable production-consumption" and "community infrastructure" as defined in ISO 37101. This can allow an evaluation of the contribution of the city to the objective "Responsible use of resources" as defined in ISO 37101.

7.7.2 Indicator requirements

The percentage of street lighting that has been renovated and newly installed in a given year shall be expressed as the number of street lighting points renovated and newly installed (numerator) in the year divided by the total number of street lighting points (denominator). The result shall then be multiplied by 100 and expressed as the percentage of street lighting that has been renovated and newly installed.

Where possible, cities should report and show separately the percentage of street lighting that is refurbished and newly installed.

	Number of street lighting points	Percentage of street lighting points
Updated		
Just installed		
Replaced		
Not maintained		
Total number		

Improvements to existing streetlight systems, such as increasing *ballast capacity*, should encompass activities aimed not only at reducing energy consumption but also at improving the energy efficiency of the streetlight system. This should include newly installed streetlights and those retrofitted to high-efficiency technology.

The light point must refer to a single source of public street lighting, such as a street light , lamppost, street lamp , or standard lamp.

SNI ISO 37122:2019**7.7.3 Data interpretation**

Cities should consider the life cycle of street lighting assets when replacing, upgrading, and installing street lighting. Cities should develop a street lighting life cycle plan to better assess the current condition of city lighting assets and identify what maintenance/materials/technologies should be used for street lighting.

7.7.4 Data sources

Data on street lighting repairs and street lighting systems must be sourced from regional government agencies or ministries responsible for street lighting inventory.

7.8. Percentage of public buildings requiring renovation/repair**7.8.1 General**

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Buildings are the largest consumers of energy in most cities. Reducing and efficient energy use can generate significant savings and can improve the stability of energy supplies. Therefore, buildings in need of renovation/repair can hinder energy efficiency, thereby contributing more to climate change and other negative externalities.

NOTE 2 This indicator reflects the issues of “Economy, sustainable production-consumption” and “Living and working environment” as defined in ISO 37101. It can enable the evaluation of the contribution of the city to the objectives of “Attractiveness” and “Responsible use of resources” as defined in ISO 37101.

7.8.2 Indicator requirements

The percentage of public buildings requiring renovation/repair will be calculated by dividing the square meters of public buildings requiring renovation/repair (numerator) by the total square meters of public buildings (denominator). The result will then be multiplied by 100 and expressed as a percentage of public buildings requiring renovation/repair.

Public buildings refer to government-owned or leased buildings that serve as offices, libraries, recreation centers, hospitals, schools, fire stations or police stations, etc.

NOTE: Building ownership (public or private) is defined differently by region and political system. The restrictive definition used here allows for global comparability across cities.

Renovations/repairs should refer to buildings, building remodeling, and general renovations of buildings to achieve better energy use, improve structural integrity, and meet safety standards. Buildings in need of renovation/repair should be assessed through a city audit and a building registration for renovation/repair. Renovations and repairs do not include demolition/relocation and replacement of buildings, but do include buildings in need of renovation.

7.8.3 Data sources

Data on public buildings requiring renovation/repair must be sourced from regional government agencies, ministries or institutions that oversee buildings and building maintenance in the city.

7.9. Percentage of buildings in the city with smart energy meters

7.9.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Smart energy meters record and display real-time energy consumption. Smart meter data can be transmitted wirelessly to a central location, giving energy providers a way to understand how and when power is used to better plan and conserve energy. Additionally, smart meter data helps consumers better understand and monitor energy usage.

NOTE 2 This indicator reflects the issues of “community infrastructure” and “Economy, sustainable production-consumption” as defined in ISO 37101. This can allow the evaluation of the contribution to the objectives of “Responsible use of resources” and “attractiveness” of the city as defined in ISO 37101.

7.9.2 Indicator Requirements

The percentage of buildings in a city with smart energy meters should be calculated by dividing the number of buildings in the city with smart energy meters (the numerator) by the total number of buildings in the city (the denominator). The result should be multiplied by 100 and expressed as a percentage of buildings in the city with smart energy meters.

Data for public, commercial and industrial buildings must be entered and listed individually.

	Number of buildings in the city with smart energy meters	Total number of buildings in the city	Percentage of buildings in the city with smart energy metering
Public buildings			
Commercial and industrial buildings			

Public buildings refer to government-owned or leased buildings that serve as city and administrative offices, libraries, recreation centers, hospitals, schools, fire stations or police stations.

NOTE 1 Building ownership (public or private) is defined differently according to region and political system. The restrictive definition used here allows for global comparability across cities.

Commercial and industrial buildings refer to buildings that have been designated by the city for commercial and industrial use.

NOTE 2 Property valuation methods may vary from one jurisdiction or country to another, including market-oriented methods, profit-oriented methods and cost-oriented methods.

Household buildings are not considered in this indicator.

For smart energy management at the household scale, see indicator 12.1.

A smart energy meter should refer to an energy meter that includes a real *-time* digital display or one available through a live online app, allowing customers to better understand their energy usage. Smart meters can also digitally transmit meter readings to energy suppliers for more accurate billing and for better energy planning and conservation by the provider.

7.9.3 Data Sources

7.10. Number of electric vehicle charging stations per registered electric vehicle

7.10.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Unlike conventional vehicles that use petrol or diesel engines, electric vehicles (EVs) are powered by electricity from batteries. Therefore, EVs emit fewer greenhouse gases and exhaust pollutants than conventional vehicles. EVs are also cheaper to operate due to minimal or zero fuel costs. However, with limited motor and battery capacity (meaning shorter driving range), electric cars require regular access to charging stations (e.g., battery chargers).

NOTE 2 This indicator reflects the issues of "Health and care in the community", "Community infrastructure" and "Living and working environment" as defined in ISO 37101. This can enable the evaluation of the contribution of the objectives to "Well-being" and "Attractiveness" and "Environmental preservation and enhancement" of the city as defined in ISO 37101.

7.10.2 Indicator Requirements

The number of electric vehicle charging stations per registered electric vehicle should be calculated by dividing the total number of electric vehicle charging stations in the city (numerator) by the total number of registered electric vehicles in the city (denominator). The result should be expressed as the number of electric vehicle charging stations per registered electric vehicle.

An electric vehicle refers to any means by which something or someone is carried or transported by means of an engine and wheels (including cars, buses, motorcycles and auto-rickshaws, but not trains) and which runs wholly or partly on a battery-powered electric motor. Electric vehicles, therefore, require "plugging in" to a power source to recharge their batteries. There are two types of electric vehicles:

1. "Hybrid" vehicles powered by a gasoline or diesel engine and electric motors;
2. "Battery electric" vehicles powered exclusively by batteries and does not require liquid fuel.

A charging station refers to publicly accessible equipment (also called “electric vehicle supply equipment” or EVSE) that supplies electrical energy to recharge electric vehicle batteries. Charging stations are often provided in municipal parking lots by electric utility companies or in retail shopping malls by private companies. Some charging stations have advanced features such as smart metering, cellular capabilities, and network connectivity.

Registered vehicle shall refer to any vehicle that has been officially registered or recorded with a government authority and that displays a vehicle registration plate and/or vehicle registration certificate.

7.10.3 Data Sources

Data for this indicator should be obtained from the urban or state agency or ministry responsible for transportation and vehicle registration in the city.

8. Environment and climate change

8.1. Percentage of buildings constructed or renovated in the last 5 years in accordance with green building principles

8.1.1 General

NOTE 1 Buildings constructed according to green building principles are substantially more sustainable. 'Green' buildings are built to higher design standards that dramatically reduce energy consumption. Green buildings can also be built or refurbished according to green building standards, which offer sustainable building benchmarks for tracking environmental performance.

NOTE 2 This indicator reflects the issues of “community infrastructure” and “Working and living environment” as defined in ISO 37101. This can allow the evaluation of the contribution to the objectives of “Responsible use of resources”, “attractiveness” and “Well-being” of the city as defined in ISO 37101.

8.1.2 Indicator Requirements

The percentage of buildings constructed or renovated in the last 5 years in accordance with green building principles will be calculated by dividing the total number of buildings constructed or renovated in the last 5 years in accordance with green building principles (numerator) by the total number of buildings in the city constructed or renovated in the last 5 years (denominator). The result will then be multiplied by 100 and expressed as the percentage of buildings constructed or renovated in the last 5 years in accordance with green building principles.

Building improvements should refer to activities aimed at not only reducing energy consumption but also improving energy efficiency and reducing the building's environmental impact. Renovations do not include the relocation and/or replacement of buildings.

Green building principles should refer to a set of guidelines and criteria by which buildings can be judged to have been constructed in accordance with "green building". Green building is the practice of creating structures and using processes that are responsible.

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environmentally and resource efficient throughout the building life cycle from placement to design, construction, operation, maintenance, renovation and deconstruction.

This practice expands and complements the classic building design concerns of economy, utility, durability, and comfort. Green buildings are also known as sustainable or high-performance buildings.

Green buildings can be buildings that are built or refurbished according to green building standards and can be classified as green buildings under standards such as BREEAM, LEED, CASBEE, HQE, BOMA BEST, BCA Green Mark, DGNB and ASGB.

Buildings do not need to be certified as green buildings, but simply need to follow green building standards during the construction process.

8.1.3 Data Sources

Data on the number of buildings constructed or renovated in the last 5 years in accordance with green building principles must be sourced from regional government agencies and ministries that oversee the construction and maintenance of buildings in the city, or oversee building permits and standards.

8.2. Number of *real-time* remote air quality monitoring stations per square kilometer (km²)

8.2.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Remotely operated *real-time* air monitoring systems can help assess the impact of climate change on the environment (e.g., air quality). Such systems can also provide direct observation, data processing and analysis, providing people with timely information about the city's air quality.

NOTE 2 This indicator reflects the issues of "Health and care in society" and "Living and working environment" as defined in ISO 37101. This can enable the evaluation of the contribution to the objectives of "Well-being", "Attractiveness" and "Environmental preservation and improvement" of the city as defined in ISO 37101.

8.2.2 Indicator Requirements

The number of real-time remote air quality monitoring stations per square kilometer (km²) should be calculated by dividing the total number of real-time remote air quality monitoring stations in the city (numerator) by the city's land area (denominator). The result should be expressed as the number of live remote air quality monitoring stations per km². Monitoring stations should refer to physical structures or devices that use specialized equipment and analytical methods to track levels of pollutants, such as fine particulate matter (PM_{2.5}), carbon dioxide (CO₂) and sulfur dioxide (SO₂).

A live remote system refers to any form of technology that provides instant information, such as mobile applications. More specifically, a remote system consists of hardware, software, data, and the people who use it. Remote systems generally include communications technology, such as the internet.

8.2.3 Data Sources

The number of *real-time* remote air quality monitoring stations must be sourced from regional government organizations or ministries that oversee city air quality.

8.3. Percentage of public buildings equipped to monitor indoor air quality

8.3.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Poor indoor air quality affects the health, comfort, and productivity of building occupants. This impact can affect a large number of occupants and especially sensitive people such as children or the elderly. To limit the health and economic consequences of poor indoor air quality, smart cities can measure and identify the sources and factors affecting indoor air quality and then propose appropriate solutions.

NOTE 2 This indicator reflects the issues of "Health and care in society" and "Working and living environment" as defined in ISO 37101. This can enable the evaluation of the contribution to the objectives of "Well-being", "Attractiveness" and "Preservation and enhancement of the environment of the city" as defined in ISO 37101.

8.3.2 Indicator Requirements

The percentage of public buildings equipped to monitor indoor air quality shall be calculated by dividing the total number of public buildings in the city equipped to monitor indoor air quality (numerator) by the total number of buildings in the city (denominator). The result shall then be multiplied by 100 and expressed as a percentage of public buildings equipped to monitor indoor air quality.

Public buildings refer to government-owned or leased buildings that serve as city and administrative offices, libraries, recreation centers, hospitals, schools, fire stations or police stations.

NOTE: Building ownership (public or private) is defined differently according to region and political system. The restrictive definition used here allows for global comparability across cities.

Indoor air quality monitoring should include primary pollutants (CO, Benzene, formaldehyde, radon, asbestos, Acetaldehyde, Toluene, Ethylbenzene, Xylenes).

8.3.3 Data Sources

Data on the number of public buildings equipped to monitor indoor air quality should be obtained from local authorities, officials, or the Ministry or Department responsible for public buildings.

9. Finance

9.1. The amount of annual revenue collected from the sharing economy as percentage of own source income

9.1.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 The sharing economy, or *peer-to-peer* sharing of access to goods and services, is a growing component of the urban economy. Incorporating this economy into existing policies allows for taxation, which supplements the city's capital budget.

NOTE 2 This indicator reflects the issue of "Economic and sustainable production and consumption" as defined in ISO 37101. It can enable the evaluation of the contribution to the objectives of "Attractiveness" and "Resilience" of cities as defined in ISO 37101.

9.1.2 Indicator requirements

The annual revenue collected from the sharing economy as a percentage of own-source revenue shall be expressed as the total amount of funds collected annually from permit fees, user fees, license fees and taxes as permitted by law or statute from sharing economy transactions (numerator) divided by the total local revenue (denominator) of the city. The result shall then be multiplied by 100 and expressed as the annual revenue collected from the sharing economy as a percentage of own-source revenue.

Total own resource income will be calculated as total income minus transfers.

The sharing economy should refer to the use of digital platforms or portals to reduce the scale of viable hiring transactions or viable participation in consumer hiring markets (i.e. 'sharing' in the sense of hiring assets) and thereby reduce the extent to which assets are underutilized for accommodation, transportation, consumer durables, labor and human capital, and intellectual property.

Broadly speaking, the sharing economy is a socio-economic ecosystem built around the sharing of human and physical resources. It encompasses the co-creation, production, distribution, trade, and consumption of goods and services by different individuals and organizations.

The sharing economy is also known as *the on-demand economy*, collaborative consumption or *peer-to-peer economy*.

9.1.3 Data sources

Tax data collected from the profit-sharing economy must be sourced from regional government agencies or relevant ministries that oversee city finances.

9.1.4 Data interpretation

The sharing economy, or *peer-to-peer* sharing of access to goods and services, is a growing component of the urban economy. Although the sharing economy remains

is a topic of debate, the inclusion of the sharing economy into existing policies allows for taxation that complements urban capital budgets.

9.2. Percentage of payments to cities paid electronically based on electronic invoice

9.2.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 The use of electronic invoicing (e-invoices) and payment transfers to municipalities increases security and reduces costs for municipalities, businesses, and their citizens. Cities that combine electronic invoicing and electronic transfers with automated accounting and control systems can experience significant productivity gains.

NOTE 2 This indicator reflects the issues of "Economy and sustainable production and consumption", "Safety and security" and "Governance, empowerment and inclusion" as defined in ISO 37101. This can enable the evaluation of the contribution to the "Attractiveness" and "Responsible Use of Resources" objectives of the city as defined in ISO 37101.

9.2.2 Indicator requirements

The percentage of payments to municipalities paid electronically based on electronic invoices (e-invoices) will be calculated as the sum of payments to municipalities made electronically based on e-invoices (numerator) divided by the total amount of payments made to municipalities (denominator). The result will then be multiplied by 100 and expressed as the percentage of payments to municipalities paid electronically based on e-invoices.

Electronic invoicing (e-invoicing) refers to the exchange of invoice documents between municipalities and businesses or citizens in electronic format. Invoices are generated directly from the relevant system, rather than simply scanned paper invoices, Word/PDF documents, or paper invoices sent to businesses or citizens. Therefore, generating electronic invoices is a natural part of the administrative work process.

Electronic payment is payment via electronic media without using cash or checks.

9.2.3 Data sources

Data on payments to the city paid electronically must be sourced from the city's economic department or from another city department responsible for payments.

9.2.4 Data interpretation

A high percentage of electronic payments is an indication that the city is highly digital.

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10. Government

10.1. Annual number of online visits to the city's open data portal per 100,000 resident

10.1.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Open data portals provide a means to increase public access to city-managed data. This creates greater transparency and enables innovation by community organizations and citizens. While many cities offer online portals, not all are equally visited.

NOTE 2 This indicator reflects the issues of "Governance, empowerment and engagement" as defined in ISO 37101. This can enable the evaluation of the contribution to the objective of "Attractiveness" of cities as defined in ISO 37101.

10.1.2 Indicator requirements

The annual number of online visits to a city's open data portal per 100,000 residents should be calculated as the total number of visits to the city's open data portal (numerator) divided by 1/100,000 of the city's total population (denominator). The result should be expressed as the annual number of online visits to the city's open data portal per 100,000 residents.

An open data portal should refer to a data portal operated by a city that provides access to open data. Open data should refer to structured, machine-readable, and freely shareable data that can be used and built upon without restriction.

Online visits will refer to individual visitors who arrive at the city's open data portal online and begin browsing and perusing the open data portal.

The visit counts all visitors, no matter how many times the same visitor visits the open data portal.

10.1.3 Data sources

Data on the number of visits to open data portals should be obtained from statistics *hosting* websites obtained from the city website administration or provided by the domain *host*.

10.2. Percentage of city services that are accessible and that can be requested online online

10.2.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Providing city services that can be requested online through digital portals benefits both citizens and local governments. Cities can provide services without fixed hours and can provide these services with reduced resource requirements. Furthermore, the use of mobile technologies, such as geotagging and photos, contributes to the efficiency and effectiveness of city services.

NOTE 2 This indicator reflects the issues of "Governance, empowerment and engagement" as defined in ISO 37101. This can enable the evaluation of the contribution to the objectives of "Social cohesion" and "Attractiveness" of the city as defined in ISO 37101.

10.2.2 Indicator requirements

The percentage of city services that are accessible and requestable online (e.g., via the internet) shall be calculated as the total number of city services offered to people and businesses through a centralized internet interface (numerator) divided by the total number of city services offered by the city (denominator). The result shall then be multiplied by 100 and expressed as a percentage of city services that are accessible and requestable online.

City services should refer to services provided by the city, including those contracted by third parties, and typically include the following areas: trash and recycling; public safety; fire; streets and traffic; regulations, violations, and enforcement; permits and licenses; planning; buildings; policies, projects, and initiatives; leasing and catering of city buildings; water and sewer; and property taxes and utilities. City services is a broad term that encompasses the many "touch points" a city has with citizens and businesses. Specifically for city services accessible online, this term may include, for example, requesting and receiving permits; assessing and collecting taxes; filing and handling complaints; and requesting information about services within the city's jurisdiction or authority.

Where possible, cities should report the percentage of users who make payments through the online fine system for traffic violations separately.

10.2.3 Data sources

An inventory of all city services offered should be taken to provide accurate percentage results. Information about city services should be sourced from the city department or agency providing the service. Figures on services available via the web or mobile should also be obtained from the city department or agency providing the service, or from the city government website administrator.

10.2.4 Data interpretation

10.3.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 The non-emergency inquiry system is a critical access point to city services. This refers to the level of response of non-emergency access points through various media including phone, app, Twitter, email, and in-person contact. Access points can be used by both citizens and businesses.

NOTE 2 This indicator reflects the issues of "Governance, empowerment and inclusion" and "Living and Working Environment" as defined in ISO 37101. This can enable the evaluation of the contribution to the objectives of "Attractiveness" and "Well-being" of the city as defined in ISO 37101.

SNI ISO 37122:2019**10.3.2 Indicator requirements**

The average response time to inquiries made through the city's non-emergency inquiry system shall be expressed as the total number of hours from initial call/form submission taken to respond to all inquiries made through the city's non-emergency system (numerator) divided by the total number of inquiries received by the city's non-emergency system (denominator). The result shall be divided by 24 and expressed as the average response time to inquiries made through the city's non-emergency inquiry system in days.

Inquiries should be from citizens and businesses that address existing issues that are economically and practically feasible to address in the short term. For example, these aren't requests for new bike lanes or parks, but rather inquiries that address urgent and timely issues, such as reporting a dead animal, requesting the planting, pruning, or removal of a tree, or filing an accessibility complaint about a city program or service.

Automatically returned receipts will not count as responses. Responses must be personalized (e.g., immediate solution, delayed resolution, or clear explanation of irrelevance).

A non-emergency inquiry system should refer to a system that citizens contact when their health, safety, or property is not in immediate danger, or when no crime is in progress. Non-emergency inquiry systems can include *hotlines*, internet-based applications (e.g., web pages, social media, mobile apps) that allow residents to submit inquiries, such as complaints about objectionable city conditions or nuisances (e.g., reporting potholes, requesting street cleaning or graffiti removal, reporting broken traffic signals) to a centralized location. The system dispatcher *then* forwards the inquiry to the appropriate city authority.

10.3.3 Data sources

Data on response times to relevant inquiries made through the city's non-emergency inquiry system should be sourced from records maintained by the non-emergency inquiry system and the relevant city department dispatched to answer the inquiry.

10.4. Average downtime of the city's information technology (IT) infrastructure**10.4.1 General**

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 In a commercial environment, the costs of downtime during a security incident – from lost sales and revenue to loss of customer confidence – can have a negative impact on the business. A comparable impact on the city's service performance/commitment can be expected.

NOTE 2 This indicator reflects the issues of "Community infrastructure" and "Safety and security" as defined in ISO 37101. This can enable the evaluation of the contribution to the objectives of "Attractiveness" and "Resilience" of cities as defined in ISO 37101.

10.4.2 Indicator requirements

The average downtime of a city's IT infrastructure during an incident should be calculated as the number of hours that the city's IT infrastructure is unavailable due to incidents (e.g., power outages, scheduled maintenance) (numerator) divided by the total number of incidents that cause IT infrastructure outages (denominator). The result should be expressed as the average downtime of the city's IT infrastructure .

An incident should include both planned and unplanned outages of a city's IT infrastructure systems (e.g., websites, payment systems). This can include planned system maintenance outages, as well as outages due to unforeseen events such as cyberattacks and power outages.

IT infrastructure shall refer to the hardware, software, networks, data centers, related facilities and equipment used to develop, test, operate, monitor, manage and/or support information technology services, for example, but not limited to, municipal data centers, server and computer computers, multi-function devices and wireless devices.

10.4.3 Data sources

Data on the outage of city IT infrastructure during incidents should be sourced from the city departments and relevant ministerial responses for IT infrastructure.

11. Health

11.1. Percentage of city population included in online integrated health files accessible to health care providers

11.1.1 General

NOTE 1 Digitization and centralization of health records enable health care providers to use the following information and its utilization of health records enable health care providers to use the following information using a holistic approach. Healthcare providers, regardless of their specialty or location, can access these individuals' health records and provide better care.

NOTE 2 Although health services are often outside the jurisdiction of city-level governments, the availability and access to primary health care is one of the characteristics of cities that can potentially impact health and is therefore an important focus area for cities. Furthermore, city health care providers face significant pressure and challenges to reduce the cost of providing health care while improving the quality of that care. Allowing health care providers to access patients' integrated health files online can help address this issue.

11.1.2 Indicator requirements

The percentage of the city population with an online integrated health file accessible to health care providers should be calculated as the total number of people with a health file.

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The number of integrated online health files accessible to all types of health care providers (numerator) is divided by the total population in the city (denominator). The result is then multiplied by 100 and expressed as a percentage of the city's population with integrated online health files accessible to health care providers.

An integrated health file should refer to a health file containing all patient health records, which are typically scattered across multiple healthcare providers, resulting in fragmented care. An integrated health file should display all medications and medical records created by both general practitioners and private practitioners. It brings together relevant information from various parts of the healthcare system, such as hospitals and family physicians, clinics, and testing centers.

Disclosure of data in the healthcare system must be done carefully and judiciously, taking into account the privacy and security measures necessary to ensure that the information will only be used for the benefit of citizens.

11.1.3 Data sources

Data on the number of people with integrated health histories accessible to health care providers should be sourced from local, city, district or provincial health providers or insurers, or relevant departments and ministries.

11.2. Annual number of medical appointments conducted remotely per 100,000 resident**11.2.1 General**

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Remote medical appointments provide a much-needed alternative to traditional in-person medical appointments. Considerations may include an aging population, decreased mobility, or limited access to transportation.

NOTE 2 This indicator reflects the issue of "Health and care in the community" as defined in ISO 37101. This can enable the evaluation of the contribution to the objectives of "Resilience", "Social cohesion" and "Well-being" of the city as defined in ISO 37101.

11.2.2 Indicator requirements

The annual number of medical appointments conducted remotely per 100,000 residents will be calculated as the total number of medical appointments conducted remotely, such as through online video services or teleconferencing (numerator) divided by 1/100,000 of the city's total population (denominator). The result should be expressed as the annual number of medical appointments conducted remotely per 100,000 residents.

A medical appointment refers to a patient's visit to a healthcare facility, where the patient discusses their health needs and concerns with one or more healthcare providers. Only formal consultations will be included.

Medical appointments conducted remotely should include those enabled by video and teleconferencing technology in accessible formats; mobile phones; remote data collection and telemonitoring equipment (e.g., cardiac monitors). Face-to-face consultations should be excluded.

11.2.3 Data sources

Data on the annual number of medical appointments conducted remotely should be sourced from the local government organization, department or ministry that oversees the health care provided to the city's residents and tracks the patterns of health care provided to its residents.

11.3. Percentage of city population with access to real -time public warning systems for air and water quality advisories

11.3.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Poor air and water quality affects human health and contributes to human mortality and morbidity in cities. Air quality warning systems provide the public with important information and advice to minimize exposure to air pollution.

Similarly, water quality alert systems inform people whether a city's water quality is suitable for drinking or other uses. Air and water quality alert systems can help reduce or mitigate the impact of pollutants on public health.

NOTE 2 This indicator reflects the issue of "Health and care in the community" as defined in ISO 37101. It can enable the evaluation of the contribution to the objectives of "Resilience, and "Well-being" of cities as defined in ISO 37101.

11.3.2 Indicator requirements

The percentage of a city's population with access to a *real-time* public warning system for air and water quality advisories shall be calculated as the number of people with access to a *real-time* public warning system for air and water quality advisories (numerator) divided by the total city population. The result shall then be multiplied by 100 and expressed as a percentage of the city's population with access to a real-time public warning system for air and water quality advisories.

A public warning system for air and water quality should refer to a system that reports and notifies the public about levels of pollutants, allergens, and particulate matter, and releases live data or data based on forecasting methods to the public. It also facilitates early identification of local air and water pollution problems and issues timely warnings about air and water pollution levels (reflecting particulate matter and pollutant levels) to the public via text messages, emails, or pre-recorded voice messages. The warning system can be in the form of a mobile app or online.

In some cases, air and water quality may be reported by two separate public warning systems, one responsible for air quality and the other for water quality. People registered with more than one public warning system will be counted only once for this indicator. Furthermore, the calculation will only include people living in the city and exclude people who would not be included in the city's total population, and therefore, those outside the city's administrative boundaries.

11.3.3 Data sources

Data on the number of people who have access to the real-time public warning system should be sourced from the relevant city department responsible for managing the warning system.

12. Housing

12.1. Percentage of households with smart energy meters

12.1.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Smart energy meters record and display energy consumption in *real time*. Smart meter data can be transmitted wirelessly to a central location, giving electricity providers the means to understand how and when power is used to better plan and conserve energy. In addition, smart meter data helps consumers better understand and monitor energy usage.

NOTE 2 This indicator reflects the issues of “Community infrastructure” and “Economy and sustainable production and consumption” as defined in ISO 37101. This can allow the evaluation of the contribution to the “Responsible use of resources” and “Attractiveness” of cities objectives as defined in ISO 37101.

12.1.2 Indicator requirements

The percentage of households with smart energy meters is calculated as the total number of households with smart energy meters (the numerator) divided by the total number of households in the city (the denominator). The result is then multiplied by 100 and expressed as the percentage of households with smart energy meters.

A smart energy meter should refer to an energy meter that includes a real-time digital display visualized online or available through a real-time online app so customers can better understand their energy usage. Smart energy meters can also digitally transmit meter readings to energy suppliers for more accurate energy billing and for better energy planning and conservation by the provider.

Where possible, the percentage of households with smart energy meters by energy type should be reported separately in the table.

	Percentage of households using smart energy meters (by energy type)
Electricity	
Gas	
District heating network	

12.1.3 Data sources

Data on smart energy meter readings should be sourced from local or regional energy providers, or relevant city departments or ministries that have local smart energy meter data.

12.2 Percentage of households with smart water meters

12.2.1 General

Those implementing this document must report these indicators in accordance with the following requirements.

NOTE 1 Smart water meters record and display water consumption in real time. Smart meter data can be transmitted wirelessly to a central location, providing water providers with the means to understand how and when water is used to better plan and conserve water. In addition, smart meter data helps consumers better understand and monitor water usage.

NOTE 2 This indicator reflects “Economic and sustainable production and consumption” and “community infrastructure” as defined in ISO 37101. It can enable the evaluation of the contribution to the “Responsible use of resources” and “Attractiveness” objectives of the city as defined in ISO 37101.

12.2.2 Indicator requirements

The percentage of households with smart water meters will be calculated as the total number of households with smart water meters (numerator) divided by the total number of households in the city (denominator). The result should be multiplied by 100 and expressed as a percentage of households with smart water meters. Smart water metering should refer to water meters that include a real-time digital display in the home with online visualization of real-time information or that are available through a real-time online application, so customers can better understand their water usage. Also, smart water meters can digitally send meter readings to the water supplier for more accurate water billing, and for better water planning and conservation by the provider.

12.2.3 Data sources

13. Population and social conditions

13.1. Percentage of public buildings accessible to people with special needs

13.1.1 General

NOTE 2 This indicator reflects the “Living together, interdependence and mutual benefit” and “Living and working environment” issues as defined in ISO 37101. It can enable the evaluation of the contribution to the objectives of “social cohesion”, “Attractiveness” and “Well-being” of the city as defined in ISO 37101.

13.1.2 Indicator requirements

The percentage of public buildings accessible to people with disabilities is calculated by dividing the number of public buildings in a city that are accessible to people with disabilities (the numerator) by the total number of public buildings in the city (the denominator). The result is then multiplied by 100 and expressed as a percentage.

The definition of accessible public buildings should be based on relevant national standards to identify, remove and prevent barriers so that people with special needs have more opportunities in everyday life.

Accessible public buildings typically include these requirements: — accessible parking spaces

— accessible main entrances — automatic

doors — adequate lighting — accessible bathrooms —

elevators to all floors

13.1.3 Data Sources

Information should be obtained from local authorities, officials, or the Ministry or Department responsible for public buildings.

13.2. Percentage of city budget allocated to providing mobility aids, devices, and assistive technologies for citizens with special needs

13.2.1 General

Those who apply this document must report these indicators in accordance with the following requirements.

NOTE 1 Ensuring a city is accessible to all its citizens and visitors promotes an equitable and inclusive society. Allocating a portion of the city budget to the provision of mobility aids, devices, and assistive technologies for citizens with special needs helps maintain the city's accessibility year after year for all its citizens and visitors and to support the autonomy (and home care) of people with special needs, including seniors.

NOTE 2 This indicator reflects the issues of “Governance, empowerment and inclusion” and “Living together, interdependent and mutually beneficial” as defined in ISO

37101. This can enable the evaluation of contributions to the “Well-being”, “Social cohesion” and “Attractiveness” objectives of cities as defined in ISO 37101.

13.2.2 Indicator Requirements

The percentage of the city budget allocated to the provision of mobility aids, devices, and assistive technologies for citizens with special needs will be calculated by the sum of the costs of providing mobility aids, devices, and assistive technologies that the city spent in a fiscal year (numerator) divided by the total city budget allocated for that particular year (denominator). The result will then be multiplied by 100 and expressed as a percentage.

13.2.3 Data Sources

Data for the amount of the city budget allocated for the provision of mobility aids, devices and assistive technologies for citizens with special needs must be sourced from the city budget and audited financial documents, or the department or ministry that oversees city expenditures for the provision of mobility aids, devices and assistive technologies for citizens with special needs.

13.3. Percentage of marked pedestrian crossings equipped with accessible pedestrian signals

13.3.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Accessible pedestrian signals enable people with special needs to safely cross intersections and to carry out their daily activities.

NOTE 2 This indicator reflects the issues of “Living together, interdependence and mutual benefit” and “Community infrastructure” as defined in ISO 37101. It can enable the evaluation of the contribution to the objectives of “Well-being”, “Social cohesion” and “Attractiveness” of the city as defined in ISO 37101.

13.3.2 Indicator requirements

The percentage of marked pedestrian crossings equipped with accessible pedestrian signals shall be calculated as the number of marked pedestrian crossings equipped with accessible pedestrian signals (numerator) divided by the total number of marked pedestrian crossings (denominator). The result shall then be multiplied by 100 and expressed as the percentage of marked pedestrian crossings equipped with accessible pedestrian signals.

Accessible pedestrian signals shall refer to devices that communicate at intervals that a crossing is safe or unsafe to enter either using non-visual communication, typically audible or *vibrotactile* (e.g. vibration), or as a supplement to visual signals.

13.3.3 Data sources

Data on the percentage of marked pedestrian crossings equipped with accessible pedestrian signals should be sourced from the city department or ministry that oversees public right-of-way and traffic signals.

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13.4. Percentage of city budget allocated to providing programs aimed at bridging the digital divide

13.4.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 As cities experience demographic shifts, the need for age-friendly urban design and city services becomes increasingly critical. Cities need to address the consequences of these unprecedented demographic shifts through urban-friendly planning and city services. Developing programs (e.g., technology classes for senior citizens) is one way to create an environment where senior citizens, but also people with disabilities, can acquire or improve technology skills to actively participate in a technology-driven society and combat the digital divide. It also empowers citizens to become active users of new technologies.

NOTE 2 This indicator reflects the issues of “Governance, empowerment and inclusion” and “Living together, interdependence and mutual benefit” as defined in ISO 37101. It can enable the evaluation of the contribution to the “Well-being”, “social cohesion” and “Attractiveness objectives” of the city as defined in ISO 37101.

13.4.2 Indicator requirements

The percentage of a city's budget allocated to providing programs aimed at bridging the digital divide shall be calculated as the sum of the city's annual expenditures on programming designed to bridge the digital divide (the numerator) divided by the city's total annual budget (the denominator). The result shall then be multiplied by 100 and expressed as a percentage of the city's budget allocated to providing programs aimed at bridging the digital divide.

The digital divide refers to economic, educational, and social disparities in access to information and communication technologies, such as computers or the Internet. Programming designed to bridge the digital divide may include, but is not limited to, programs to help low-income and vulnerable communities (e.g., children, youth, and seniors) gain access to, and learn about, new technologies by providing hardware, software, and Internet access.

13.4.3 Data sources

Data on city budgets allocated to provide programs aimed at bridging the digital divide must be sourced from the city budget.

14. Recreation

14.1. Percentage of public recreation services that can be booked online

14.1.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Online recreation bookings offer increased accessibility and awareness for the public, as well as a data source for public recreation participation.

NOTE 2 This indicator reflects the "Living and working environment" issue as defined in ISO 37101. This can enable the evaluation of the contribution to the "Well-being" objective of the city as defined in ISO 37101.

14.1.2 Indicator Requirements

The percentage of public recreation services that can be booked online will be calculated by dividing the number of public recreation services that can be booked online (numerator) by the total number of public recreation services offered by a city (denominator).

The result will then be multiplied by 100 and expressed as a percentage of public recreation services that can be booked online.

Recreational services refer to those that operate facilities or provide services that enable people to participate in sports or recreational activities or pursue pleasures, hobbies, and leisure interests. Recreational services may include municipalities providing public recreation spaces, broadly defined as land and buildings open to the public for recreation, such as swimming, sports, skating facilities, and fitness centers. Municipal recreation services may also include municipally managed programs, camping, and facility rentals.

14.1.3 Data Sources

The percentage of public recreation services that can be booked online must be sourced from the relevant city department or ministry that oversees public recreation, or the department responsible for online administration.

15. Security

15.1. Percentage of city area covered by digital surveillance cameras

15.1.1 General

Those who apply this document must report these indicators in accordance with the following requirements.

NOTE 1 The presence of surveillance cameras is a deterrent to crime and other offenses. When incidents do occur, video surveillance offers an accurate representation of events, as well as key information for solving crimes and other offenses. Digital cameras are more reliable than film, and they have higher capacities, better image quality, and create files that are easy to distribute and difficult to alter.

NOTE 2 This indicator reflects the "Safety and security" issue as defined in ISO 37101. It can allow evaluation of the contribution to the "Well-being" and "Attractiveness" objectives of the city as defined in ISO 37101.

15.1.2 Indicator Requirements

The percentage of the city area covered by digital surveillance cameras shall be calculated by dividing the total area of the city land covered by digital video surveillance cameras in square kilometers (numerator) by the total area of the city land (denominator). The result shall then be multiplied by 100 and expressed as a percentage of the city area covered by digital surveillance cameras.

Digital surveillance cameras, sometimes referred to as Internet protocol (IP) cameras, should refer to video cameras that can send and receive data over a computer network, while the other party sends the feed to a digital video recorder (DVR) (i.e. disk/USB)

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push). The measurement of the range is determined by the technical specifications of the system used.

This indicator should include digital surveillance that is accessible to the city, such as digital video surveillance that is directly accessible to the city or law enforcement, without the permission of, or request from, the private camera owner.

When using a digital surveillance camera system, one must consider how the data/images are used, as well as the privacy and security measures necessary to ensure the safety of citizens.

15.1.3 Data Sources

Data on the percentage of city area covered by digital surveillance cameras should be sourced from local law enforcement and safety departments, ministries or agencies.

15.1.4 Data Interpretation

Data on the percentage of city area covered by digital surveillance cameras should be sourced from local law enforcement and safety departments, ministries or agencies.

16. Solid Waste

16.1. Percentage of waste disposal centers (containers) equipped with telemetering

16.1.1 General

Those who apply this document must report these indicators in accordance with the following requirements.

Drop-off center (container) collection is defined as follows: it is a collection method in which the community provides the population with a network of waste bins distributed throughout the city and available to all citizens with open access. Residents who do not have proper waste bins must dispose of sorted materials at community-managed sites, such as a container next to a public road, a parking lot or commercial center, or a district collection point.

Collection trucks will collect waste (containers) from allocated points in the city. When trucks are equipped with telemetering to optimize their rotation depending on bin filling, this can be considered as access for citizens to deliver collections with telemetering.

NOTE 1 Many cities have had to restrict traffic within the city and simplify the organization of waste collection. In addition, many cities have narrow and substandard roads that provide only limited access to households and neighborhoods. In cities in less developed countries, roads and lanes are not always accessible to garbage trucks for collection.

Developing telemetered waste drop-off centers (where residents bring their waste) is a local solution that can help cities achieve their goals of reducing traffic, addressing limited access, and streamlining waste collection and disposal. Telemetering aids in the optimization and efficiency of waste collection by alerting collection trucks to the level of waste currently stored in containers at the drop-off center.

NOTE 2 This indicator reflects the issues of “Community infrastructure”, “Economy and sustainable production and consumption” and “Mobility” as defined in ISO 37101. This can allow the evaluation of the contribution to the objectives of “Responsible use of resources” and “Environmental preservation and enhancement” of the city as defined in ISO 37101.

16.1.2 Indicator Requirements

The percentage of waste disposal centers (containers) equipped with telemetering shall be calculated by dividing the number of waste disposal centers (containers) equipped with telemetering devices (numerator) by the total number of waste disposal centers (containers) in the city (denominator). The result shall then be multiplied by 100 and expressed as a percentage of waste disposal centers (containers) equipped with telemetering.

Waste disposal centers (containers) must be located where people drop off waste according to sorting criteria. *Drop-off* centers can, for example, be located near public roads or in parking facilities. People using drop-off centers generally don't have their own trash bins. Collection trucks will collect the waste at the drop-off centers (containers).

Waste disposal centers (containers) and waste collection vehicles are equipped with telemetering to optimize waste collection rounds based on information about the waste bins.

Telemetering refers to measurements using means that allow the measurements to be interpreted at a distance from the primary detector. A characteristic of telemetering is the translation of the means, which includes provisions for converting the measurement into a representative quantity of another type that can be easily transmitted for measurement at a distance. If a waste disposal center (container) is equipped with telemetering, the amount of waste in the waste disposal center (container) can be remotely transmitted to a waste collection truck.

16.1.3 Data Sources

Data on the percentage of waste disposal centers equipped with telemetering should be sourced from the city department that oversees waste/garbage disposal centers.

16.2. Percentage of the city population that has door-to-door waste collection with individual monitoring of the amount of household waste

16.2.1 General

Those who apply this document must report these indicators in accordance with the following requirements.

NOTE 1 Individual monitoring of household waste volumes provides valuable information for citizens and cities. Understanding the weight of household waste can help optimize waste collection and reduce costs. Furthermore, telemetering reduces road traffic congestion by comparing the number of vehicles with the actual amount of waste to be collected. The benefits include smoother traffic flow with consequent reductions in Greenhouse Gas (GHG) emissions, better collection cycle design, and better allocation of human resources with corresponding savings.

NOTE 2 This indicator reflects the issues of “Community infrastructure”, “Economy and sustainable production and consumption” and “Mobility” as defined in ISO 37101. This can

16.2.2 Indicator Requirements

allows for the evaluation of contributions to the objectives of "Responsible use of resources" and "Environmental preservation and enhancement" of the city as defined in ISO 37101.

The percentage of the city's population that has door-to-door waste collection with individual monitoring of household waste volumes shall be calculated by dividing the number of people living in the city with door-to-door household waste collection equipped with monitoring devices (numerator) by the total city population (denominator). The result shall then be multiplied by 100 and expressed as a percentage of the city's population that has door-to-door waste collection with individual monitoring of household waste volumes.

16.2.3 Data Sources

Data on the percentage of the city population that has door-to-door waste collection equipped with individual monitoring (including telemetering data) should be sourced from the relevant agency that oversees waste collection services and door-to-door waste collection.

16.3. Percentage of the total amount of waste in the city that is used to produce energy

16.3.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Wastes containing significant organic matter can be a source of energy either directly by recovering heat from energy from waste plants (incinerators) or by generating energy from waste digestion or other new technologies using this energy for cogeneration, biogas production for injection in the gas network, or for fuel production.

In a context where energy consumption from fossil fuels must be reduced for sustainable development purposes, it is advisable to use these heat, electricity, gas, or other fuel sources for other city-wide services (e.g., swimming pool heating, fuel for the city's vehicle fleet, energy sales to local industries). This is also a way for cities to achieve a degree of energy independence.

NOTE 2 This indicator reflects the issues of "Community Infrastructure", "Economy and sustainable production and consumption" as defined in ISO 37101. It can enable the evaluation of the contribution to the objectives of "Responsible use of resources" and "Environmental preservation and enhancement" and "Resilience" of cities as defined in ISO 37101.

16.3.2 Indicator Requirements

The percentage of the total amount of waste in a city that is used to generate energy is calculated by dividing the total amount of waste used to generate energy (the numerator) by the total amount of waste generated in the city (the denominator). The result is then multiplied by 100 and expressed as a percentage of the total amount of waste in the city that is used to generate energy.

The energy generated from a waste treatment plant must be expressed in GJ per year.
The total amount of waste used to produce energy must refer to waste processing that has a positive net energy production rate.

16.3.3 Data Sources

Data on the amount of waste in a city can be obtained from the ISO 37120 indicator "solid waste collection per capita of the city" multiplied by the city population.

Data on the total amount of waste in a city used to generate energy should be sourced from the local utility, or the relevant city department that oversees waste treatment and related energy generation.

16.4. Percentage of the total amount of plastic waste recycled in the city

16.4.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Plastic waste is a global environmental problem. To prevent the spread of plastic in the environment, the best solution is to limit plastic production and promote plastic recycling. Considering the potential ecological impact of microplastics on water and oceans, cities can promote plastic recycling within their territories. This requires monitoring plastic production and encouraging the increased use of recycled plastic in other products.

NOTE 2 This indicator reflects the issues of "Community infrastructure", "Economy and sustainable production and consumption" as defined in ISO 37101. It can enable the evaluation of the contribution to the "Responsible use of resources" and "Environmental preservation and enhancement" objectives of the city as defined in ISO 37101.

NOTE 3 During the procurement phase, communities may consider the percentage of recycled plastic incorporated into the product as a selective criterion.

16.4.2 Indicator Requirements

The percentage of total plastic waste recycled is calculated by dividing the total amount of plastic leaving the sorting and recycling plant (the numerator) by the total amount of plastic in the market within the city limits (the denominator). The result is then multiplied by 100 and expressed as a percentage of the total amount of recycled plastic in the city.

SNI ISO 37122:2019**16.4.3 Data Sources**

Data on the amount of plastic released from waste sorting plants in cities must be sourced from local utilities, relevant regional government organizations that oversee waste processing. Plastic data in urban markets should be sourced from commercial activities, industrial activities and waste collection utilities for household consumption or from the plastics industry.

16.5. Percentage of public waste bins that are public waste bins that function with sensors**16.5.1 General**

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Solid waste management and monitoring require urgent attention in all cities. Sensor-activated solutions for public waste bins are one way cities can improve waste monitoring and collection. Sensor-activated bins can lead to optimal route planning and waste collection scheduling, potentially leading to significant cost reductions in solid waste collection.

NOTE 2 This indicator reflects the issues of "Community infrastructure" and "Economy and sustainable production and consumption" as defined in ISO 37101. This can enable the evaluation of the contribution to the objective of responsible use of resources and the objective of "Preservation and enhancement of the environment" of the city as defined in ISO 37101.

16.5.2 Indicator Requirements

The percentage of sensor-enabled public waste bins for public waste bins will be calculated by dividing the number of sensor-enabled public waste bins (numerator) by the total number of public waste bins in the city (denominator). The result will then be multiplied by 100 and expressed as a percentage of public waste bins that are sensor-enabled public waste bins.

Public trash bins refer to trash bins provided by the city that are located in public spaces, such as on streets and in public parks.

Sensor-activated public waste bins shall refer to public waste bins that are equipped with a sensor, or sensors, that monitor waste levels and that are part of a larger network of sensor-activated waste bins connected via a telecommunications network that generates data and allows remote monitoring of fill levels.

16.5.3 Data Sources

Data on public waste bins should be obtained from the relevant regional government organization responsible for municipal waste or primary municipal waste companies/firms.

16.6. Percentage of municipal electrical and electronic waste recycled**16.6.1 General**

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 With the increasing popularity of mobile phones, computers, televisions, and other electronic devices, it is increasingly important that cities ensure that electronic waste (or e-waste) undergoes environmentally sound management at the end of its useful life. E-waste recycling programs help keep electronic devices out of landfills and recover useful resources.

NOTE 2 This indicator reflects the issues of “Community infrastructure” and “Economy and sustainable production and consumption” as defined in ISO 37101. This can allow the evaluation of the contribution to the objectives of “Responsible use of resources” and “Preservation and enhancement of the environment” of the city as defined in ISO 37101.

16.6.2 Indicator Requirements

The percentage of municipal electrical and electronic waste recycled shall be calculated by dividing the total amount of municipal electrical and electronic waste recycled in tonnes (numerator) by the total amount of electrical and electronic waste generated in the city in tonnes (denominator). The result will then be multiplied by 100 and expressed as a percentage of the city's electrical and electronic waste that is recycled.

Electrical and electronic waste, commonly referred to as *e-waste*, refers to electronics such as computers, printers or fax machines, televisions or computer displays, audio-video equipment (including DVD players, VCRs, speakers, and portable digital music players), mobile phones and electronic gaming equipment.

This indicator should include electrical and electronic waste produced by all property classes: residential, commercial, industrial and public buildings.

16.6.3 Data Sources

Data on e-waste should be sourced from the city department responsible for municipal solid waste or the municipal solid waste company/firm.

17. Sports and culture

17.1. Number of online reservations for cultural facilities per 100,000 inhabitants

17.1.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Cultural facilities and cultural/sporting events play a vital role in connecting people and in building more cohesive and open societies. Digitizing access to cultural institutions helps increase the availability of cultural resources to a wider audience.

NOTE 2 This indicator reflects the issues of “Education and capacity building” and “Culture and community identity” as defined in ISO 37101. This can enable the evaluation of the contribution to the objectives of “Social cohesion” and “Well-being” of the city as defined in ISO 37101.

SNI ISO 37122:2019**17.1.2 Indicator Requirements**

The number of online bookings for cultural facilities per 100,000 inhabitants should be calculated by the number of online bookings for cultural facilities (numerator) divided by 1/100,000 of the total city population (denominator).

Cultural facilities shall refer to public or non-profit institutions within the city engaged in the cultural, intellectual, scientific, environmental, educational, sporting or artistic enrichment of the people residing in the city. "Cultural facilities" shall include, without limitation, aquariums, botanical societies, historical societies, land conservation organizations, libraries, museums, performing arts associations or societies, scientific societies, wildlife conservation organizations, sports facilities (i.e., indoor and outdoor arenas, fields) and *zoological societies*. "Cultural facilities" shall not include educational institutions (e.g., schools) or institutions primarily engaged in religious or sectarian activities.

17.1.3 Data Sources

Data on the number of online bookings for cultural facilities should be sourced from the relevant ministries and departments responsible for cultural facilities and centers.

17.1.4 Data Interpretation

This indicator measures how online ordering systems can increase the availability of cultural resources to a wider audience.

17.2. Percentage of city cultural records that have been digitized**17.2.1 General**

Those who apply this document must report these indicators in accordance with the following requirements.

NOTE 1 The process of digital preservation, or digitization, is a formal effort to ensure that digital information, such as digital data, is managed to ensure continued access and usability. Digital preservation of cultural records is one form of digital preservation that ensures cultural artifacts are preserved for future users. Furthermore, digital preservation connects and gives people broader access to heritage materials, which helps stimulate an innovative information society. The digitization of a city's cultural records contributes to the conservation and preservation of heritage and scientific resources; it creates new educational opportunities; can be used to encourage tourism; and provides a way to increase access by citizens to their heritage.

NOTE 2 This indicator reflects the issue of "Culture and community identity" as defined in ISO 37101. This can enable the evaluation of the contribution to the objectives of "Resilience" and "Attractiveness" of cities as defined in ISO 37101.

17.2.2 Indicator Requirements

The percentage of a city's cultural records that have been digitized will be calculated as the number of digitized city cultural records (numerator) divided by the total number of city cultural records (denominator). The result should be multiplied by 100 and expressed as a percentage of digitized cultural records.

Digitization refers to the conversion of traditional analog materials such as books, maps, and other physical items (paper records) into electronic and digital copies.

The cultural record should refer to the tangible cultural heritage, or the legacy of physical artifacts and texts, of the city and include movable cultural heritage (paintings, sculptures, coins, manuscripts); immovable cultural heritage (e.g., monuments, archaeological sites); and underwater cultural heritage (shipwrecks, ruins, and underwater cities), where applicable. A city's cultural record is likely to be extensive, especially when considering the cultural record in all its myriad forms. Therefore, this indicator will only include tangible cultural heritage owned and/or managed by the city to ensure data accessibility. It excludes privately owned components of the city's cultural record.

17.2.3 Data Sources

Data on cultural records of a city should be sourced from relevant city archives, or relevant departments and ministries.

17.3. Number of public library books and e-book titles per 100,000 population

17.3.1 General

NOTE 1 Libraries help educate the general public, in addition to providing civic spaces for interaction.

They can be considered local gateways to knowledge, and provide “the basic conditions for lifelong learning, independent decision-making and the cultural development of individuals and social groups.” Those who implement this document must report these indicators in accordance with the following requirements. Ultimately, as stated in the UNESCO Public Library Manifesto, “public libraries [can be considered] a vital force for education, culture and information, and as essential agents for promoting peace and spiritual well-being through the minds of men and women.”

Electronic books (*e-books*) have become popular among the public due to their easy accessibility, allowing citizens to more easily pursue lifelong learning and cultural development, and be exposed to a wealth of information.

Furthermore, the availability of e-books indicates the level of digitalization of community libraries, as well as the ease of access to books through library websites. Furthermore, *e-books* are relatively environmentally friendly and promote sustainability, as they require less paper and labor to produce, and they do not require shelf space.

has grown significantly in popularity and prevalence in the publishing industry.

17.3.2 Indicator Requirements

The number of library books and *e-book* titles per 100,000 residents should be calculated by dividing the total number of library book titles and the total number of library e-book titles (numerator) by 1/100,000 of the city's total population (denominator). The result should be expressed as the number of library book titles per 100,000 residents.

Public libraries should refer to libraries within the administrative boundaries of a city.

The proportion of *e-books* in the total number of library books must be determined.

Cities should include in this indicator the number of library books and *e-book* titles available to their citizens from public libraries. Cities should count each book title.

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that are available and accessible in public libraries within city limits, and include physical books/monographs and *e-books*, which include books/monographs that have been digitized and available for reading on computers or other electronic devices. A public library is any library that has an organized collection of printed or other library materials, or a combination thereof, is supported in whole or in part by public funds and has a set schedule on which staff services are available for

general.

	Number of public library book titles
<i>E-books</i>	
Book	

17.3.3 Data Sources

Data on the number of library book titles should be sourced from the local library, library board or relevant city department, or through the ministry.

17.4. Percentage of the city's population that are public library users active**17.4.1 General**

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Libraries help educate the general public, in addition to providing civic spaces for interaction. The number of active library users is a measure of the reach and effectiveness of local libraries which provide "the basic conditions for lifelong learning, independent decision-making and the cultural development of individuals and social groups".

NOTE 2 This indicator reflects the issue of "Education and capacity development" as defined in ISO 37101. This can allow the evaluation of the contribution to the objectives of "Social cohesion" and "Well-being" of the city as defined in ISO 37101.

17.4.2 Indicator Requirements

The percentage of a city's population that is a public library user shall be calculated by dividing the total number of city residents who are active library users as citizens registered as members of a public library or as those using library services (numerator) by the total city population (denominator). The result shall then be multiplied by 100 and expressed as a percentage of the city's population that is an active public library user.

Active public library users should refer to registered public library members, or people who have a user account at a public library who frequently use library services with at least one transaction per month, such as accessing the online library/database to download articles or *e-books* that require library user privileges, or the process of checking out books from the library. Active library users recorded in this indicator will only include users who live in the city. The city must count each user

active libraries with public libraries within city limits. Users from outside the administrative boundaries should be excluded.

Urban areas must count the active library users within the urban area. A public library is a library that has a collection of printed materials or other forms or a combination thereof, is partially or wholly supported by public funding and has a regular schedule where staff are available to serve the public.

17.4.3 Data Sources

Data on the number of active library users should be sourced from the local library, library board or relevant city department, or through the ministry.

17.4.4 Data Interpretation

A large number of active library users indicates that city libraries meet the needs of residents and that libraries help to educate residents.

18. Telecommunications

18.1. Percentage of city population with access to *broadband* speed adequate

18.1.1 General

NOTE 1 Adequate *broadband* speed helps enable individuals to exercise their right to freedom of opinion and expression, and promotes social progress. This document assesses progress on these indicators in accordance with the following requirements of the United Nations, and provides citizens with the opportunity to explore and retrieve information available on the *World Wide Web*.

18.1.2 Indicator Requirements

The percentage of a city's population with access to adequate *broadband* speed should be calculated by dividing the total number of people in the city with access to adequate broadband speed (numerator) by the total population of the city (denominator).

The total will then be multiplied by 100 and expressed as a percentage of the city's population with access to adequate broadband speed.

Broadband refers to the data transmission capacity associated with a specific transmission speed and the provision of high-speed Internet access. Broadband supports applications such as web browsing, IP TV video services, and so on. Broadly speaking, broadband infrastructure is the underlying communications infrastructure used to enable the provision of broadband services, namely Internet access at a specific speed/bandwidth.

Adequate broadband speed refers to a network with a speed of at least 256 kbit/s in both directions, uploading and downloading. This speed is sufficient for browsing the internet and sending email. Adequate speed should take into account the potential network needs of both the service provider and the service user.

18.1.3 Data Sources

Data on the number of people with access to sufficiently fast broadband data *should* be sourced from local broadband service providers, relevant departments or ministries, or companies that oversee broadband services.

18.2. Percentage of city area under white zone / dead spot / not covered by telecommunication connectivity

18.2.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Telecommunications enables not only unhindered communication, but also access to services such as the Internet. Therefore, white zones and dead spots are obstacles to communication and access to basic services.

NOTE 2 This indicator reflects the issue of "Community infrastructure" as defined in ISO 37101. This can allow the evaluation of the contribution to the objectives of "Social cohesion" and "Attractiveness" of the city as defined in ISO 37101.

18.2.2 Indicator Requirements

The percentage of city area under white zone / dead spot / not covered by telecommunication connectivity shall be calculated as the total land area of the city classified in this way under white zone / dead spot / not covered by telecommunication connectivity in square kilometers (numerator) divided by the total land area of the city in square kilometers (denominator). The result will then be multiplied by 100 and expressed as a percentage of the city area under the white zone / dead spot / not covered by telecommunications connectivity.

18.2.3 Data Sources

Data on city areas under the white zone / dead spots / not covered by telecommunications connectivity must be sourced from local Internet service providers, or regional government organizations or relevant ministries that oversee the development of telecommunications infrastructure.

18.3. Percentage of city area covered by provided Internet connectivity by city

18.3.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Public Internet connections allow people who do not have mobile data plans or regular Internet access to connect to the Internet, allowing them to take advantage of the economic and social benefits the Internet can offer. In addition, publicly accessible Internet can help enable cities to passively track users for future planning purposes.

NOTE 2 This indicator reflects the issue of "Community infrastructure" as defined in ISO 37101. This can allow the evaluation of the contribution to the objectives of "Social cohesion" and "Attractiveness" of the city as defined in ISO 37101.

18.3.2 Indicator Requirements

The percentage of urban area covered by city-provided Internet connectivity shall be calculated as the total area of urban land served with Internet connectivity in square kilometers (numerator) divided by the total area of the city in square kilometers (denominator). The total is then multiplied by 100 and expressed as a percentage of the urban area covered by city-provided Internet connectivity.

Urban Internet connectivity should refer to Internet connectivity services provided by the city or a third-party provider licensed by the city to the public, and should be accessible to anyone (visitors or residents) within the city limits. Urban Internet services should include free, partially paid, or fully paid connectivity.

Publicly available internet connectivity is determined by location, not by *router*. For example, if there are multiple *routers* in a park, the park is considered a single location. Public places include, but are not limited to, parks, outdoor spaces, buildings, transportation routes, and transportation hubs and stations.

18.3.3 Data Sources

Data on urban areas covered by city-provided Internet connectivity should be sourced from the regional government organization responsible for information technology and managing the ratio of publicly accessible internet, or can be estimated using GIS tools.

19. Transportation

19.1.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

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NOTE 1 The prominence and growth of online citizenship tools has created a culture of real-time sharing of citizenship data , including online traffic alerts and information. This data can be crowdsourced .

Geospatial data from mobile data, or collected through sensors or cameras installed by road and transportation authorities. The application of this technology allows authorities to efficiently plan for future conditions and allows users to travel effectively on public and toll roads.

NOTE 2 This indicator reflects the issues of "Community Infrastructure" and "Mobility" as defined in ISO 37101. This can allow the evaluation of the contribution to the objectives of "Attractiveness", "Well-being" and "Environmental preservation and enhancement" of the city as defined in ISO 37101.

19.1.2 Indicator Requirements

The percentage of public roads and toll roads covered by *real-time* online traffic alerts and information shall be calculated by dividing the number of kilometers of public roads and toll roads within the city limits covered by real-time online traffic alerts and information (numerator) by the total number of kilometers of public roads and toll roads within the city limits (denominator). The result shall then be multiplied by 100 and expressed as the percentage of public roads and toll roads covered by real-time online traffic alerts and information.

Public roads and toll roads shall refer to all local roads, streets, and major and minor arterial roads within the city.

A real-time information system refers to an information processing system that will respond to externally generated input within a limited and defined timeframe. In the context of online traffic alerts and information, "real -time" refers to traffic information that is instantly available and reflects traffic conditions at any given moment.

19.1.3 Data Sources

Data on public roads and toll roads covered by real-time online traffic alerts and information must be sourced from relevant regional government agencies, or agencies that manage and disseminate online content related to traffic in a particular region.

19.2. Number of users of shared economy transportation per 100,000 population**19.2.1 General**

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Cities are increasingly leveraging the sharing economy to complement existing mobility needs. The extent to which policymakers and planners are aware of the number of sharing economy users in a city will allow for the development of better plans and reconfiguration of the city's transportation system to accommodate these changes.

NOTE 2 This indicator reflects the issues of "Mobility" and "Living together, interdependence and mutual benefit" as defined in ISO 37101. It can enable the evaluation of the contribution to the objective of "Social cohesion" and the objectives of "Attractiveness" and "Well-being" of the city as defined in ISO 37101.

19.2.2 Indicator Requirements

The number of shared economy transportation users per 100,000 residents should be calculated by dividing the total number of users actively using shared economy transportation (the numerator) by 1/100,000 of the city's total population (the denominator). The result should be expressed as the number of shared economy transportation users per 100,000 residents.

The sharing economy should refer to any form of economic activity in which platform providers and customers leverage information technology to exchange goods and services that are often previously underutilized (see ISO/IWA 27:2017). The sharing economy comprises marketplaces and platforms that enable individuals and organizations to buy and sell goods and services directly with each other, and to rent, share, or lend goods or assets on a short-term or time-share basis. Sharing economy transportation for this indicator should refer to any mode of transportation in which individuals can utilize assets owned by other individuals or organizations, such as ride-sharing and car-sharing services.

19.2.3 Data Sources

Data on the number of shared economy transportation users must be sourced from relevant regional government agencies or shared economy transportation service companies. Given the tensions between city authorities and shared economy transportation providers in many jurisdictions, it can be difficult for cities to access the necessary data.

19.2.4 Data Interpretation

While the shared economy is a reality for many cities, it has the potential for negative impacts on public transportation. There's growing evidence that shared rides may be cannibalizing passengers from public transportation.

19.3. Percentage of registered vehicles in the city that are low-cost vehicles emission

19.3.1 General

19.3.2 Indicator Requirements

The percentage of vehicles registered in a city that are low-emission vehicles will be calculated by dividing the total number of registered and approved low-emission vehicles registered in the city (numerator) by the total number of registered vehicles in the city (denominator). The result should be multiplied by 100 and expressed as a percentage of vehicles registered in the city that are low-emission vehicles.

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Low-emission vehicles shall refer to vehicles that have low emission levels and may include electric, hybrid, and hydrogen-powered vehicles. Low-emission vehicles shall be certified to the appropriate exhaust emission standards and shall meet any other specific requirements applicable to conventional or clean-fuel vehicles and their fuels.

NOTE Air quality is measured in ISO 37120:2018, 8.1 and 8.2.

19.3.3 Data Sources

The number of registered and approved low-emission vehicles must be sourced from regional government agencies, or agencies that oversee vehicle registration.

19.4. Number of bicycles available through city-provided bicycle sharing services per 100,000 inhabitants

19.4.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Bike sharing or bike-sharing schemes are services that provide bicycles for shared use by individuals for short periods. Typically, individuals can borrow and return bicycles at different locations. Bike sharing encourages increased bicycle use in cities by reducing common barriers to cycling, including cost, bicycle theft, and repairs. Bike sharing provides an alternative to conventional modes of transportation such as public transport or private cars. This indicator provides cities with a measure of bicycle availability in bike-sharing systems.

NOTE 2 This indicator reflects the issues of "Community infrastructure" and "Mobility" as defined in ISO 37101. It can allow the evaluation of the contribution to the objectives of "Social cohesion" and "Attractiveness" and "Well-being" of the city as defined in ISO 37101.

19.4.2 Indicator Requirements

The number of bicycles available through city-provided bike-sharing services per 100,000 population should be calculated by dividing the total number of bicycles available through city-provided bike-sharing services (numerator) by 1/100,000 of the city's total population (denominator). The result should be expressed as the number of bicycles available through city-provided bike-sharing services per 100,000 population.

Bike-sharing services refer to bike-sharing systems where bicycles are available through self-service or user-operated docking stations located throughout a city, where bicycles can be rented as needed. Users must be able to rent and return bicycles to any docking station within the bike-sharing system.

Government-provided bike-sharing services shall refer to bike-sharing services funded and operated by a city. This shall also include bike-sharing services officially operated through a cooperative agreement or memorandum of understanding with a city government, such as a public-private partnership.

19.4.3 Data Sources

Data on the number of bicycles available through bicycle sharing services in a city should be sourced from the relevant local government organizations that monitor and/or collect data on bicycle sharing.

19.5. Percentage of public transportation routes equipped with publicly accessible *real-time* systems

19.5.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

19.5.2 Indicator Requirements

The percentage of public transport lines equipped with publicly accessible *real-time* systems shall be calculated by dividing the number of public transport lines equipped with publicly accessible real-time systems to provide operational information directly to people (numerator) by the total number of public transport lines within the city limits (denominator). The result shall then be multiplied by 100 and expressed as a percentage of public transport lines equipped with publicly accessible real-time systems.

A public transport route refers to a portion of the public transport network where public transport vehicles depart and arrive from two points on the public transport network in a single, continuous trip, following a schedule with travel times and stops that should be the same at all times. The calculation of this indicator should include rail and road-based public transport, cable cars, and waterway transport. Public transport routes should be distinguished from public transport routes when calculating this indicator, so a public transport route may include multiple public transport routes.

A publicly accessible direct system should refer to any information processing system that responds to externally generated input within a limited and defined period, and that provides instant information to users. In the context of public transportation lines, direct systems provide timely information on transit usage and current ridership on public transportation lines, so that routes and modes of transportation can be planned in the most efficient manner. The information provided should not be limited to users of a particular transportation line; but should be publicly available to allow access for all citizens.

19.5.3 Data Sources

Data on the percentage of public transport routes equipped with real *-time* ICT-based systems should be taken from the relevant regional government agencies that supervise public transport and monitor traffic.

SNI ISO 37122:2019**19.6. Percentage of city public transport services covered by the integrated payment system****19.6.1 General**

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Integrated payment systems encourage multiple modes of transportation across modes of transportation such as buses, LRT, subways and trains, and reduce the need for public transportation users to stop and pay at multiple transit points during a single trip. Integrated payment systems for public transportation users are not limited to a specific route or mode of transportation, but cover all types of public transportation modes.

NOTE 2 This indicator reflects the issues of "Community Infrastructure" and "Mobility" as defined in ISO 37101. This can allow the evaluation of the contribution to the objectives of "Social cohesion" and "Attractiveness" and "Well-being" of the city as defined in ISO 37101.

19.6.2 Indicator Requirements

The percentage of city public transport services covered by the unified payment system shall be calculated by dividing the number of city public transport services connected by the unified payment system (numerator) by the total number of city public transport services (denominator). The result shall then be multiplied by 100 and expressed as a percentage of city public transport services covered by the unified payment system.

Public transportation services refer to travel services provided locally by a city that allow a number of people to travel together along a designated route. The most common public transportation vehicles that make up a public transportation network can include vehicles provided and/or managed by the city, such as buses, boats, subways, trains, shared bicycles, and shared cars.

A unified payment system should refer to an integrated mobility payment system that allows transit users to plan, book, and pay for multiple transit modes to get them from point A to point B. A unified payment system should include ICT-based user/interface technologies such as smart cards or mobile tickets, and a unified pricing structure, so that transit users do not have to pay at multiple transit points when making a single trip.

19.6.3 Data sources

Data on city public transportation services covered by the integrated payment system must be sourced from the relevant regional government organization responsible for the city's transit system.

19.7. Percentage of public parking spaces equipped with payment systems electronic**19.7.1 General**

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Electronic payment systems offer the public a more convenient payment method because they do not rely on cash or checks, and reduce the time spent in queues. Electronic payment systems also create opportunities for intelligent pricing, depending on the time of day or frequency of use.

NOTE 2 as This indicator reflects the issues of "Community Infrastructure" and "Mobility" defined in ISO 37101. May allow evaluation of contributions to the objectives of "Attractiveness" and "Well-being" of cities as defined in ISO 37101.

19.7.2 Indicator Requirements

The percentage of public parking spaces equipped with electronic payment systems will be calculated by dividing the number of public parking spaces equipped with electronic payment systems as the payment method (numerator) by the total number of public parking spaces in the city (denominator). The result will then be multiplied by 100 and expressed as the percentage of public parking spaces equipped with electronic payment systems. Public parking spaces will be calculated based on capacity, and street parking will be calculated based on individual paid spaces.

Electronic payment systems will refer to ways of making transactions or paying for goods and services through electronic media without using checks or cash, such as credit cards or online or mobile applications.

19.7.3 Data sources

The percentage of public parking spaces equipped with electronic payment systems must be sourced from the local government organization that oversees public parking, as well as any organization (public or private) that oversees electronic payment systems in the city that are relevant to public parking.

19.8. Percentage of public parking spaces equipped with a parking availability system *real-time parking*

19.8.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Direct systems help distribute information about parking availability, hours of operation, pricing guidelines, and accessibility options. In addition, direct systems help people more efficiently identify available public parking spaces, thereby helping reduce fuel use and vehicle emissions in the process.

NOTE 2 This indicator reflects the issues of "Community infrastructure" and "Mobility" as defined in ISO 37101. It can allow the evaluation of the contribution to the objectives of "Attractiveness", "Well-being" and "Environmental preservation and enhancement" of the city as defined in ISO 37101.

19.8.2 Indicator Requirements

The percentage of public parking spaces equipped with a direct availability system should be calculated by dividing the number of public parking spaces equipped with a direct availability system (numerator) by the total number of public parking spaces in the city (denominator). The result will then be multiplied by 100 and expressed as a percentage of public parking spaces with a direct availability system.

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Public parking spaces should be calculated based on their capacity (i.e. number of public spaces), and street parking should be calculated based on individual paid spaces.

A real-time availability system for public parking spaces should include any form of technology that provides instant information, such as through mobile and/or online applications, about the availability of public parking spaces (e.g., the number of public parking spaces available).

19.8.3 Data Sources

Data on the number of public parking spaces with a real *-time* availability system must come from regional government organizations that supervise public parking.

19.9. Percentage of intelligent traffic lights**19.9.1 General**

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Intelligent traffic lights help optimally control the flow of vehicles and pedestrians through roads and intersections, thereby improving mobility and reducing transportation fuel consumption. They can also be used to inform emergency responders moving quickly within cities of optimal routes.

NOTE 2 Automobile technology has begun to implement *anti-idling systems*, which can work more efficiently if they can communicate with intelligent/smart traffic lights to predict light changes and reduce emissions.

NOTE 3 as This indicator reflects the issues of “Community Infrastructure” and “Mobility” defined in ISO 37101. This can enable the evaluation of contributions to the objectives of “Well-being” and “Environmental preservation and enhancement” of the city as defined in ISO 37101.

19.9.2 Indicator Requirements

The percentage of smart traffic lights should be calculated by the number of smart traffic lights in the city (numerator) divided by the total number of traffic lights in the city (denominator). The result will then be multiplied by 100 and expressed as a percentage of the smart traffic light.

Smart traffic lights should refer to traffic light systems that utilize a combination of lights, sensors, and other information and communication technologies, along with algorithms, to control the flow of vehicular and pedestrian traffic.

Multiple traffic lights at the same intersection for traffic going in the same direction should be counted as a single traffic light.

19.9.3 Data sources

Data on the percentage of smart traffic lights should be sourced from relevant regional government organizations that oversee transportation and street lights.

19.10. City area mapped by real *-time* interactive street map as a percentage of the total city area

19.10.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Live interactive street maps provide up-to-date information for people traveling through the city, or planning to travel in and around the city. This allows people to more efficiently plan their travel times and routes, as well as identify access points that accommodate people with special needs.

NOTE 2 This indicator reflects the issues of "Community Infrastructure" and "Mobility" as defined in ISO 37101. This can enable the evaluation of the contribution to the objectives of "Well-being" and "Environmental preservation and enhancement" of the city as defined in ISO 37101.

19.10.2 Indicator Requirements

The city area mapped by the live interactive street map as a percentage of the total city area will be calculated by dividing the total city area mapped by the live interactive street map (numerator) by the total land area of the city (denominator). The result will then be multiplied by 100 and expressed as the city area mapped by the live interactive street map as a percentage of the total city area.

An interactive street map refers to a street map generated by a geographic information system (GIS) and containing location labels that respond digitally and directly to a *mouse*, web cursor, or touchpad. These labels correspond to the locations of businesses or buildings that are accessible to people with disabilities.

Real *-time* information should be updated instantly on interactive street maps to reflect recent changes in an area, such as road construction or business relocations. Street maps should include pedestrian and sidewalk networks and the city's public transportation network.

19.10.3 Data Sources

19.11. Percentage of registered vehicles in the city that are autonomous *vehicles*

19.11.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 *Autonomous* vehicles could reduce traffic fatalities by eliminating crashes caused by human error, which could be the most significant advance in the history of automotive safety. This could be achieved by shifting the focus from minimizing post-crash injuries to preventing crashes altogether.

NOTE 2 This indicator reflects the issue of “Mobility” as defined in ISO 37101. This can allow the evaluation of the contribution to the objective of “Preservation and enhancement of the city environment” as defined in ISO 37101.

19.11.2 Indicator Requirements

The percentage of vehicles registered in a city that are autonomous *should* be calculated by dividing the total number of autonomous vehicles registered in the city (the numerator) by the total number of vehicles registered in the city (the denominator). The result should be multiplied by 100 and expressed as the percentage of vehicles registered in the city that are autonomous.

Autonomous vehicles should refer to vehicles that can drive themselves (i.e. do not require a human driver).

19.11.3 Data Sources

The number of autonomous vehicles registered in a city must be sourced from the regional apparatus organization or agency that oversees vehicle registration.

19.12. Percentage of public transport routes with city-provided and/or managed internet connectivity for commuters

19.12.1 General

Those who apply this document must report these indicators in accordance with the following requirements.

NOTE 1 Public Internet connections allow people who do not have mobile data plans or regular Internet access to connect to the Internet, allowing them to take advantage of the economic and social benefits the Internet offers. In addition, publicly accessible Internet can help cities passively track users for future planning.

NOTE 2 This indicator reflects the issues of “Mobility” and “Community Infrastructure” as defined in ISO 37101. It can allow the evaluation of the contribution to the objectives of “Attractiveness” and “Well-being” of cities as defined in ISO 37101.

19.12.2 Indicator Requirements

The percentage of public transit routes with city-provided and/or managed connectivity for commuters shall be calculated by dividing the number of kilometers of public transit routes in the city with city-provided and/or managed Internet connectivity for commuters (numerator) by the total number of kilometers of public transit routes in the city (denominator). The result shall be multiplied by 100 and expressed as a percentage of public transit routes with city-provided and/or managed Internet connectivity for commuters.

Municipally provided and/or managed Internet connectivity shall refer to Internet connectivity services provided and/or managed by the city or a third-party provider under license by the city to the public, and shall be accessible to anyone (visitor or resident) within the city limits.

19.12.3 Data Sources

Data regarding the percentage of public transit routes with city-provided and/or managed internet connectivity for commuters should be sourced from the relevant public transit company and/or city public transportation department.

19.13. Percentage of roads suitable for autonomous driving systems

19.13.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 The suitability of roads for automated driving systems requires a database that accurately defines the road (road type, number of lanes, traffic data) as well as infrastructure that ensures real -time localization of autonomous vehicles (e.g. availability of communication network infrastructure [GNSS, Wi-Fi, 5g]).

NOTE 2 This indicator reflects the issues of "Mobility" and "Community Infrastructure" as defined in ISO 37101. This can allow the evaluation of the contribution to the objective of "Attractiveness" of cities as defined in ISO 37101.

19.13.2 Indicator Requirements

The percentage of roads suitable for autonomous driving systems is calculated by dividing the number of road kilometers suitable for autonomous driving systems (the numerator) by the total number of road kilometers (the denominator). The result is then multiplied by 100 and expressed as the percentage of roads suitable for autonomous driving systems.

19.13.3 Data Sources

Data regarding the percentage of roads suitable for autonomous driving systems must be sourced from relevant regional government organizations.

19.14. Percentage of city bus fleet that is motorized

19.14.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 The deployment of motor-driven public transport vehicles instead of engine-driven ones helps cities reduce operating costs and vehicle exhaust emissions, while providing an environmentally friendly mode of transport to public transport users. In addition, motor-driven public transport vehicles reduce noise and vibrations emanating from the engine system, thereby improving passenger safety and comfort.

NOTE 2 This indicator reflects the issues of "Mobility" and "Community Infrastructure" as defined in ISO 37101. This can allow the evaluation of the contribution to the objective of "Attractiveness" of cities as defined in ISO 37101.

19.14.2 Indicator Requirements

The percentage of the city bus fleet that is motor-driven should be calculated by the number of buses in the city bus fleet that are motor-driven (numerator) divided by the total number of buses

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in the city bus fleet (denominator). The result will then be multiplied by 100 and expressed as a percentage of the city bus fleet that is motorized.

Motor-driven vehicles shall refer to buses that are propelled by a motorized system (not an engine-driven system that burns or consumes fuel to perform mechanical work), and that use motors driven by electricity (magnetic force), air, hydraulic pressure, heat, photons, electrons, or ultrasound. Motors do not change the chemical composition of their energy source. Motor-driven systems include, but are not limited to, battery-powered systems containing fuel cells, and exclude biogas and internal combustion engine-driven systems that require diesel.

NOTE: Batteries have a life cycle of more than 10 years, while the life cycle of fuel cell batteries has not been confirmed due to insufficient usage experience.

19.14.3 Data Sources

Data on the city bus fleet must be sourced from the relevant regional government organizations responsible for the city's transit system.

19.14.4 Data Interpretation

With regard to sustainability, the energy sources supplying the city bus fleet must be taken into account. See ISO 37120:2018, 7.2 for characterizing a city's energy mix.

20. Urban/local agriculture and food resilience**20.1. Annual percentage of city budget spent on agricultural initiatives
urban****20.1.1 General**

Those implementing this document must report this indicator in accordance with the following requirements.

NOTE 1 Urban agriculture makes an important contribution to household food security, especially in times of food crisis or shortage. Locally produced food requires shorter supply chains and less transportation and refrigeration, and thus can help conserve energy, water, and other resources.

NOTE 2 This indicator reflects the issues of "Health and care in the community" and "Economy and sustainable production and consumption" as defined in ISO 37101. This can allow the evaluation of the contribution to the objective of "Environmental preservation and enhancement" of the city as defined in ISO 37101.

20.1.2 Indicator Requirements

The annual percentage of a city's budget spent on urban agriculture initiatives should be calculated by dividing the total amount of the city's budget spent on urban agriculture initiatives for a given year (the numerator) by the total city budget for that same year (the denominator). The result will then be multiplied by 100 and expressed as an annual percentage of the city's budget spent on urban agriculture initiatives.

Urban agriculture refers to the growing of crops and food products from a variety of plants (grains, root crops, vegetables, mushrooms, fruits). Urban agriculture also includes trees managed for fruit production and small-scale aquaculture.

In many cities, raising animals (e.g., poultry, rabbits, goats, sheep, cattle, pigs, wild boar) within city limits is prohibited by law. Urban agriculture initiatives or programs should address any activities related to the definition of urban agriculture above or support urban agriculture activities, such as city grants available to urban agriculture producers and businesses that can support the development of innovative technologies for urban agriculture (e.g., mobile apps for monitoring crop yields) or simply provide resources to urban agriculture producers and businesses to support general operations.

20.1.3 Data Sources

Data on the amount of city budget spent on urban agriculture initiatives should be sourced from audited financial reports from the city budget or relevant regional government organizations that oversee finances.

20.2. Total annual collected municipal food waste sent to processing facilities for composting per capita (in tonnes)

20.2.1 General

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Although food and organic matter are essential for life and healthy soil, a significant amount of food and organic waste ends up in municipal waste streams, for example, in landfills or incinerators. It is recognized within cities and around the world that food and organic waste is a growing problem, and that current waste management practices are unsustainable. There are environmental consequences to sending food and organic matter to landfill. The environmental benefits of recycling and composting food waste can be significant. Composting converts food waste into usable products such as fertilizer, which can then be used in agriculture and food production, increasing food productivity and promoting smart, sustainable growth.

NOTE 2 This indicator reflects the issue of "Economic and sustainable production and consumption" as defined in ISO 37101. It can enable the evaluation of the contribution to the objectives of "Responsible use of resources" and "Resilience" of cities as defined in ISO 37101.

20.2.2 Indicator Requirements

The total annual municipal waste collected and sent to processing facilities for composting per capita (in tons) shall be calculated by dividing the total amount of food waste (household and commercial) collected in tons (numerator) by the total population of the city (denominator). The result shall be expressed as the total annual municipal food waste collected and sent to processing facilities for composting per capita in tons.

Composting refers to a natural biological process carried out under controlled aerobic conditions (requiring oxygen). In this process, various microorganisms, including bacteria and fungi, break down organic matter into simpler substances and convert the waste into a usable organic soil amendment or mulch by providing adequate aeration, moisture, particle size, fertilizer, and lime.

The effectiveness of the composting process depends on the environmental conditions present in the composting system, namely oxygen, temperature, humidity, disturbance of materials, organic materials and size.

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and microbial population activity. Thus, processing facilities for composting should refer to facilities that carry out composting.

Food waste refers to discarded and unconsumed food recovered from the food supply chain. This can occur throughout the food supply chain, from primary production to end-user consumption (residential and commercial). Food waste is recognized as a distinct subset of food loss because the drivers that generate it, and the solutions to it, differ from those that cause food loss.

20.2.3 Data Sources

Data on municipal food waste collected should be sourced from the relevant local government organizations responsible for waste collection, recycling, sanitation and/or composting services.

20.2.4 Data Interpretation

A city with a total of municipal food waste collected annually sent to processing facilities for composting per capita (in tons) is indicative of a city that is diverting and reducing the amount of waste that needs to be disposed of and, in turn, mitigating the environmental impacts associated with municipal solid waste. It also indicates a city that is converting food waste into useful products for agriculture and improving soil for food production.

20.3. Percentage of urban land area covered by the online food supplier mapping system**20.3.1 General**

Those who implement this document must report these indicators in accordance with the following requirements.

NOTE 1 Maps showing food suppliers in a city help connect residents with food sources. Food maps also provide baseline data on the status of access to nutritious food supplies and assets, allowing cities to take stock of their food resources.

NOTE 2 This indicator reflects the issues of "Health and care in the community" and "Economy and sustainable production and consumption" as defined in ISO 37101. This can allow the evaluation of the contribution to the objective of "Resilience" of cities as defined in ISO 37101.

20.3.2 Indicator Requirements

The percentage of urban land area covered by the online food supplier mapping system will be calculated by dividing the total land area covered by the online food supplier mapping system (numerator) by the total land area of the city (denominator). The result will then be multiplied by 100 and expressed as a percentage of the urban land area covered by the online food supplier mapping system.

Food supplier mapping systems should be based on information technology used by food and public health agencies to map, visualize, and analyze the distribution of food resources. Online food supplier mapping systems should be based on publicly accessible online food supplier mapping systems.

Food suppliers should refer to retail food sources, such as food retailers (e.g., restaurants, grocery stores, and convenience stores that offer food) and farms that have food available to residents that residents can purchase.

20.3.3 Data Sources

Data on the online food supplier mapping system should be sourced from the city department/ministry responsible for public health and health and wellness programs and/or GIS mapping application development.

21. Urban planning

21.1. Number of city residents per 100,000 inhabitants per year involved in the process planning

21.1.1 General

Those implementing this document must report these indicators in accordance with the following requirements.

NOTE 1 Citizen engagement is a key attribute of effective planning and policymaking. Successful citizen engagement enhances this process because citizens have input and influence on city government plans.

NOTE 2 This indicator reflects the issues of “Governance, empowerment and engagement” as defined in ISO 37101. This can enable the evaluation of the contribution to the objective of “social cohesion” of the city as defined in ISO 37101.

21.1.2 Indicator requirements

The number of city residents per 100,000 population per year involved in the planning process is calculated as the total number of residents participating or involved in the planning process on an annual basis (numerator) divided by 1/100,000 of the city's total population (denominator). The result will be expressed as the annual number of citizens involved in the planning process per 100,000 population.

The planning process must refer to official plans and other city plans.

The definition of citizen engagement includes direct attendance or involvement in events such as community consultations, public hearings, pop-up town hall sessions and other participatory practices,

For example, online hearings and webinars. Citizen engagement can also include virtual attendance or engagement through social media or formal engagement tools such as online surveys or handouts.

Cities should report the means of engagement, where possible.

Ways of involvement	Total population
Personally	
Online	

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Online engagement will be assessed through comments, likes, dislikes via social media or formal engagement tools.

21.1.3 Data sources

Data on citizen engagement should be sourced from attendance records of engagement planning processes, hearings, and events (both online and in-person) often recorded in planning and policy reports as a prerequisite for approval.

21.2. Percentage of building permits submitted through the electronic submission system

21.2.1 General

Those implementing this document must report these indicators in accordance with the following requirements.

NOTE 1 Building permit applications and approval processes can hinder development feasibility and profitability. Providing the option to submit building permit applications electronically may help expedite the permitting process by eliminating the need for city staff to perform routine data entry and allowing applicants to submit building permits more quickly.

NOTE 2 This indicator reflects the issues of "Governance, empowerment and engagement" as defined in ISO 37101. This can enable the evaluation of the contribution to the objective of "Attractiveness" of cities as defined in ISO 37101.

21.2.2 Indicator requirements

The percentage of building permits submitted through the electronic submission system is calculated as the number of building permits submitted through the electronic submission system (numerator) divided by the total number of building permits submitted through the electronic submission system and the direct manual system (e.g., paper applications). The result will be multiplied by 100 and expressed as the percentage of building permits submitted through the electronic submission system.

An electronic submission system should refer to an online system that allows applicants to complete a web-based application form and submit building permits electronically. An electronic submission system also allows applicants to upload supporting documents directly online.

21.2.3 Data sources

Data on the number of building permits submitted through the electronic submission system should be sourced from the relevant regional government agencies that oversee building permit applications.

21.3. Average time for building permit approval (days)

21.3.1 General

Those implementing this document must report these indicators in accordance with the following requirements.

NOTE 1 The development application and building permit approval process can hinder development feasibility and profitability. This indicator allows cities to compare

their development applications and buildings allow for approval time with other cities to improve their internal processes.

NOTE 2 This indicator reflects "Governance, empowerment and involvement" and "Economic and sustainable production and consumption issues" as defined in ISO 37101. It can enable the evaluation of the contribution to the objectives "Responsible use of resources" and "Attractiveness" of cities as defined in ISO 37101.

21.3.2 Indicator requirements

The average time to building permit approval should be calculated as the number of days from start to completion of a building permit (numerator) divided by the total number of building permits (denominator). The result should be expressed as the average time to building permit approval in days.

Building permit approvals should include permits for new commercial buildings, commercial building renovations and non-residential construction projects, as well as large residential projects and small residential projects in, for example, detached, semi-detached and *townhouse* construction and renovation project.

21.3.3 Data sources

Data on the average time for building permit approval should be sourced from the relevant city government that oversees building permit approvals.

21.3.4 Data interpretation

Cities with relatively low average times for building permit approval may have more efficient permit approval systems in place. However, it should be noted that direct comparisons between cities may be difficult when considering differences in the regulatory environment in which building permit approvals occur and the potential for more stringent or more stringent building permit requirements.

21.4. Percentage of the city's population living in medium to high population densities tall

21.4.1 General

NOTE 1 Population density is a fundamental condition of cities and it affects how they function. Urban planners advocate higher population densities for the widely held theory that cities operate more efficiently when residents live in denser urban environments. Higher population densities can contribute to smart growth, given that other aspects, such as car dependence, are less of a problem. Growth is "smart" because it is intended to be sustainable and long-lasting, and not solely dependent on cars.

NOTE 2 This indicator reflects the issue of "Living and working environment" as defined in ISO 37101. It can allow the evaluation of the contribution to the objectives of "Social cohesion" and "Attractiveness" and "Well-being" of the city as defined in ISO 37101.

SNI ISO 37122:2019**21.4.2 Indicator requirements**

The percentage of the urban population living in medium to high urban density should be calculated as the number of people living in areas with medium to high population density (numerator) divided by the total population of the city (denominator).

The result will then be multiplied by 100 and expressed as the percentage of the city's population living in medium to high population densities.

Cities must establish and report the range used for their population calculations ranging from medium to high population density levels.

21.4.3 Data sources

Population density data must be sourced from relevant regional government agencies that oversee urban planning and population statistics.

22. Wastewater**22.1. Percentage of treated wastewater reused****22.1.1 General**

Those implementing this document must report these indicators in accordance with the following requirements.

NOTE 1 Wastewater reuse is a means of conserving water in areas where scarcity is increasing and water shortages are likely. It is a solution consistent with the principles of a circular economy that helps address climate change and adaptation challenges.

This is also a way to prevent the discharge of untreated wastewater into the environment.

NOTE 2 This indicator reflects the "Community infrastructure", "Economic and sustainable production and consumption issues" and "Biodiversity and ecosystem services" as defined in ISO 37101. It can allow an evaluation of the contribution to the "Responsible use of resources", "Environmental preservation and enhancement" and "Resilience" objectives of cities as defined in ISO 37101.

22.1.2 Indicator requirements

The percentage of treated wastewater reused shall be calculated as the total annual volume of treated wastewater reused (numerator) divided by the total annual volume of treated wastewater (denominator). The result shall be multiplied by 100 and expressed as a percentage of treated wastewater reused.

Treated wastewater reuse shall refer to wastewater reused after secondary biological treatment ("controlled" reuse) or following conventional tertiary treatment (filtration, UV disinfection, chlorination, ozonation), or high-quality treatment after membrane treatment (MBR, ultrafiltration, ultrafiltration/microfiltration *reverse osmosis/ reverse osmosis*) for irrigated agriculture, urban irrigation (green areas) or other more noble uses such as water recycling and groundwater recharge (see ISO 24511).

22.1.3 Data sources

Data on the percentage of reused wastewater should be sourced from local government agencies, ministries, or agencies responsible for wastewater management and wastewater networks. Data can also be sourced from local utility providers, if applicable.

22.1.4 Data interpretation

Data should be analyzed in light of local water scarcity. In cities where water scarcity is not a pressing issue, other water reuse techniques, such as rainwater harvesting, may be more appropriate.

22.2. Percentage of *biosolids* reused (dry matter mass)

22.2.1 General

Those implementing this document must report these indicators in accordance with the following requirements.

NOTE 1 *Biosolids* may have significant mineral content (e.g., N, P), oligoelements, and organic matter that can be reused either for agricultural fertilization and soil improvement, or for the heating value of energy-from-waste plants or digestion facilities to produce biomethane that can be reused for gas injection or fuel production. *Biosolids* reuse is an important component of the circular economy, helping to reduce the environmental disposal or disposal of *biosolids*. Several types of *biosolids* reuse

can help mitigate the expected decline in mineral resources such as phosphorus. Therefore, the production of new phosphorus resources, such as *struvite*, will be needed in the future.

NOTE 2 This indicator reflects "Community infrastructure", "Economic and sustainable production and consumption issues" and "Biodiversity and ecosystem services" as defined in ISO 37101. It can allow an evaluation of the contribution to the objectives "Responsible use of resources", "Environmental preservation and enhancement" and "Resilience" of cities as defined in ISO 37101.

22.2.2 Indicator requirements

The percentage of *biosolids* reused shall be calculated as the total annual amount of *biosolids* reused in dry matter mass (numerator) divided by the total annual amount of *biosolids* produced and measured at the site outlet in the city in dry matter mass (denominator). The result shall then be multiplied by 100 and expressed as the percentage of *biosolids* reused in dry matter mass.

NOTE The quantity of *biosolids* measured at the site outlet is expressed in tonnes of dry matter (including additives). The quantity reused per year includes all uses except landfilling and incineration without heat recovery.

Biosolids should refer to the residue obtained after wastewater or sanitation treatment. *Biosolids* characteristics vary from source to source, depending on the type of initial waste and the type of treatment applied. Solid waste and sand from screening are not included in this definition.

Biosolids considered in the calculation of this indicator may come from:

- handling of uncollected rainwater; – *night soil* (soil containing human waste);

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– urban wastewater collection systems; – urban wastewater treatment plants; – treating industrial wastewater similar to urban wastewater; – water supply treatment plants.

However, hazardous sludge from industry is not included.

22.2.3 Data sources

Data on the annual amount of *biosolids* reused and the total annual amount of *biosolids* generated in the city should be sourced from the relevant local government agencies, ministries, or agencies responsible for solid waste management, wastewater, and sewage systems. Data can also be sourced from local utility providers, if applicable.

22.3. Energy derived from wastewater as a percentage of total consumption city energy**22.3.1 General**

Those implementing this document must report these indicators in accordance with the following requirements.

NOTE 1 Wastewater containing significant organic matter can be a source of energy producing energy from wastewater digestion or biosolids or other new technologies that use this energy for cogeneration, biomethane production for injection in the gas network, or for fuel production, recovering heat from wastewater in the wastewater network.

With the ongoing need to reduce energy consumption from fossil fuels, it is advantageous for cities to use these heat, electricity, gas, or other fuels to facilitate the delivery of other services within the city (e.g., swimming pool heating, fuel for the city's vehicle fleet, energy sales to local industries). Where fossil fuel tariffs exist and are unprofitable, cities can achieve a degree of energy independence through wastewater utilization.

NOTE 2 This indicator reflects the issues of "Societal infrastructure" and "Economic and sustainable production and consumption" as defined in ISO 37101. It can enable the evaluation of contributions to the objective "Responsible use of resources",

"Environmental preservation and enhancement" and "Resilience" of cities as defined in ISO 37101.

22.3.2 Indicator requirements

The energy derived from wastewater as a percentage of the city's total energy consumption is calculated as the sum of the total annual energy derived from the wastewater network and wastewater treatment plants (numerator) divided by the city's total energy consumption (denominator). The result is then multiplied by 100 and expressed as the energy derived from wastewater as a percentage of the city's total energy consumption.

Energy comes from the wastewater network and wastewater treatment plants, and the total energy consumption of the city, will be expressed in GJ per year.

22.3.3 Data sources

The total energy consumption data for a city can be obtained from the ISO 37120 indicator 'Energy use per capita multiplied by the city's population. Data on the amount of energy coming from

from wastewater must be sourced from the relevant regional apparatus organization or wastewater utility.

22.4. Percentage of total urban wastewater used to generate energy

22.4.1 General

Those implementing this document must report these indicators in accordance with the following requirements.

NOTE 1 Wastewater containing significant organic matter can be a source of energy producing energy from the digestion of wastewater or biosolids or other new technologies that use this energy for cogeneration, biomethane production for injection in the gas network, or for fuel production, or by recovering heat from wastewater in the wastewater network.

With the ongoing need to reduce energy consumption from fossil fuels, it is advantageous for cities to use these heat, electricity, gas, or other fuels to facilitate the delivery of other services within the city (e.g., swimming pool heating, fuel for the city's vehicle fleet, energy sales to local industries). If fossil fuel tariffs exist and are unprofitable, cities can achieve a degree of energy independence through wastewater utilization.

NOTE 2 This indicator reflects the issues of "Societal infrastructure" and "Economic and sustainable production and consumption" as defined in ISO 37101. It can enable the evaluation of contributions to the objective "Responsible use of resources",

"Environmental preservation and enhancement" and "Resilience" of cities as defined in ISO 37101.

22.4.2 Indicator requirements

The percentage of the total amount of wastewater in a city used to generate energy is calculated as the total amount of wastewater used to generate energy (numerator) divided by the total amount of wastewater in the city (denominator). The result is then multiplied by 100 and expressed as a percentage of the total amount of wastewater in the city used to generate energy.

The energy generated from a wastewater network or treatment plant must be expressed in gigajoules (GJ) per year.

22.4.3 Data sources

Data on the total amount of wastewater in the city and the total amount of wastewater in the city used to generate energy must be sourced from the local utility, or the relevant local government organization that oversees wastewater treatment and related energy generation.

22.5. Percentage of wastewater pipe network monitored by data sensor system *real-time tracking*

22.5.1 General

Those implementing this document must report these indicators in accordance with the following requirements.

NOTE 1 Equipping municipal wastewater pipe networks with sensor-based technology allows for continuous measurement of effluent levels in the network, detection of discharges into the sewers

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stormwater overflows, and calculations of flow and volume discharges to the environment and their potential cost reductions. Furthermore, sensor-based systems enable remote management and operation of wastewater and stormwater networks, detecting problems and implementing quick and efficient solutions.

NOTE 2 This indicator reflects the issues of "Community infrastructure" and "Economy and sustainable production and consumption" as defined in ISO 37101. It can enable the evaluation of the contribution to the objectives of "Responsible use of resources", "Environmental preservation and enhancement" and "Resilience" of cities as defined in ISO 37101.

22.5.2 Indicator requirements

The percentage of the wastewater pipe network monitored by the real *-time* data tracking sensor system shall be calculated as the length of the wastewater pipe network monitored by the real-time data tracking sensor system in kilometers (numerator) divided by the total length of the wastewater pipe network in kilometers (denominator). The result shall then be multiplied by 100 and expressed as the percentage of the wastewater pipe network monitored by the real-time data tracking sensor system.

A sensor system should refer to a network of devices (e.g., sensors) that detect and respond to some type of input from the physical environment. A live data tracking sensor system for pipeline wastewater should refer to a sensor system that provides instantaneous data on pipeline wastewater.

22.5.3 Data sources

Data on wastewater pipe networks must be sourced from the relevant regional government organization responsible for wastewater, or public utility organizations.

22.5.4 Data interpretation

Since this indicator relates to digitalization tools, technological advancements in other areas, such as network planning, construction, and renovation, should be taken into account. The ultimate goal of a "smart" city should be to achieve sustainability goals, not simply the aimless use of digitalization tools.

23. Water**23.1. Percentage of drinking water tracked by real *-time* water quality monitoring stations****23.1.1 General**

Those implementing this document must report these indicators in accordance with the following requirements.

NOTE 1 *Real-time* ICT-based systems for monitoring drinking water quality can help inform urban residents about drinking water quality and reduce the health impacts of degraded drinking water. ICT-based systems also provide direct observation, enabling immediate data processing and analysis of water quality information.

NOTE 2 This indicator reflects the issues of "Community infrastructure", "Economy and sustainable production and consumption" and "Health and care in the community" as defined

in ISO 37101. This can allow the evaluation of contributions to the objectives of "Responsible use of resources", "Resilience" and "Attractiveness" of cities defined in ISO 37101.

23.1.2 Indicator requirements

Percentage of drinking water tracked by real *-time* water quality monitoring stations is calculated as the amount of drinking water that has undergone direct water quality monitoring by water quality monitoring stations in the city (numerator) divided by the total amount of drinking water distributed in the city (denominator). The result must be multiplied by 100 and expressed as a percentage of drinking water tracked by direct water quality monitoring stations.

A monitoring station refers to a physical structure or device that uses specialized equipment and analytical methods to track the levels of urban drinking water pollutants. Tracking will include more than one measurement point along the network and cannot be limited to network entry points.

Live systems will refer to any form of technology that provides instant information such as mobile applications .

23.1.3 Data sources

The quantity of drinking water under water quality monitoring is determined by real *-time water quality monitoring*. The stations must be sourced from relevant regional government agencies that monitor urban drinking water quality.

23.1.4 Data interpretation

Since this indicator relates to digitalization tools, technological advancements in other areas, such as network planning, construction, and renovation, should be taken into account. The ultimate goal of a "smart" city should be to achieve sustainability goals, not simply the aimless use of digitalization tools.

23.2. Number of real *-time* environmental water quality monitoring stations per 100,000 population

23.2.1 General

Those implementing this document must report these indicators in accordance with the following requirements.

NOTE 1 *Real-time* systems for monitoring environmental water quality can help reduce the impacts of climate change on the environment and its aquatic ecosystems. Using ICT-based systems in environmental water monitoring can provide direct observation, providing cities and their citizens with timely information about water quality.

NOTE 2 This indicator reflects the "Community Infrastructure", "Living and working environment", "Biodiversity and ecosystems" and "Health and care in the community" issues as defined in ISO 37101. It can allow the evaluation of the contribution to the objectives of "Attractiveness", "Resilience" and "Environmental preservation and enhancement" of the city as defined in ISO 37101.

23.2.2 Indicator requirements

The number of *real-time* environmental water quality monitoring stations per 100,000 population should be calculated as the total number of environmental water quality monitoring stations.

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directly in the city (numerator) divided by 1/100 000 of the total city population (denominator).
The results are expressed as the number of direct environmental water quality monitoring stations per 100,000 population.

Environmental water refers to water in rivers or wetlands that is beneficial to the environment, for example water that is set aside in storage areas such as reservoirs and dams, and that is managed for plants and animals.

A monitoring station refers to a physical structure or device that uses specialized equipment and analytical methods to track environmental water pollution levels.

A live system refers to any form of ICT-based technology or system (such as a mobile app) that provides instant information. More specifically, an ICT system consists of hardware, software, data, and the people who use it. ICT systems typically include communications technology, such as the Internet. It's important to note that ICT and computers are not the same thing—a computer is a piece of hardware that is often part of an ICT system.

23.2.3 Data sources

The number of *real-time* ICT-based environmental water quality monitoring stations should be taken from the relevant regional apparatus organizations that monitor the water quality of the city's natural water network and urban environment.

23.3. Percentage of city water distribution network monitored by smart water systems**23.3.1 General**

Those implementing this document must report these indicators in accordance with the following requirements.

Cities must consider the pressures of residential, commercial, and industrial demand on water supplies to manage water supplies efficiently and effectively. Cities must also manage water consumption and distribution more efficiently. Cities, public water utilities, and industrial water users manage various components of water infrastructure through various methods, such as monitoring, control, and data acquisition (SCADA) systems, sensors, and meters. A smart water system is an integrated approach to managing urban water use, consisting of a network of sensors and meters that provide information on water consumption and leakage in the distribution network.

NOTE This indicator reflects the issues of "Community infrastructure" and "Economy and sustainable production and consumption" and "Health and care in the community" as defined in ISO 37101. It can allow the evaluation of the contribution to the objectives of "Responsible use of resources", "Attractiveness" and "Resilience" of cities as defined in ISO 37101.

23.3.2 Indicator requirements

The percentage of the urban water distribution network monitored by the smart water system shall be calculated as the length of the water distribution network covered by the smart water system in kilometers (numerator) divided by the total length of the water distribution network in kilometers (denominator).

The result will then be multiplied by 100 and expressed as a percentage of the water distribution network monitored by the urban smart water system.

A smart water system should encompass a network of sensors and meters that enable municipalities and utility providers to remotely monitor and diagnose problems within the water system. It also provides the ability to prioritize and manage maintenance issues, using data to optimize all aspects of the water system's pipelines. Tracking should include more than one measurement point along the network and cannot be limited to the network entry point.

23.3.3 Data sources

Data on water system networks and smart water systems must be sourced from city, district and provincial water managers or regional government agencies or relevant ministries/agencies that store local water system network data.

23.3.4 Data interpretation

Since this indicator relates to digitalization tools, technological advancements in other areas, such as network planning, construction, and renovation, should be taken into account. The ultimate goal of a "smart" city should be to achieve sustainability goals, not simply the aimless use of digitalization tools.

23.4. Percentage of urban buildings with smart water meters

23.4.1 General

Those implementing this document must report these indicators in accordance with the following requirements.

NOTE 1 Smart water meters record and display water consumption in real-time.

Smart meter data can be transmitted wirelessly to a central location, giving water providers the ability to understand how and when water is used, allowing them to better plan and conserve water. Furthermore, smart meter data helps consumers better understand and monitor their water usage.

NOTE 2 This indicator reflects the issues of "Community infrastructure" and "Economy and sustainable production and consumption" as defined in ISO 37101. It can enable the evaluation of the contribution to the objectives of "responsible use of resources" and "Attractiveness" of cities as defined in ISO 37101.

23.4.2 Indicator requirements

The percentage of buildings in a city with smart water meters will be calculated as the number of buildings in the city with smart water meters (numerator) multiplied by the total number of buildings in the city (denominator). This should be multiplied by 100 and expressed as a percentage of buildings in the city with smart water meters.

Data for public and commercial buildings must be entered and listed individually.

	Number of buildings in the city with smart water meters	The total number of buildings in the city	Percentage of buildings in the city with smart water meters
Public buildings			
Commercial and industrial buildings			

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Public buildings refer to government-owned or leased buildings that serve as city and administrative offices, libraries, recreation centers, hospitals, schools, fire stations or police stations.

NOTE 1 Building ownership (public or private) is defined differently by region and political system. The restrictive definition used here allows for global comparability across cities.

Commercial and industrial properties refer to properties designated by the city for commercial and industrial use.

NOTE 2 Property valuation methods may vary from one jurisdiction or country to another, including market-oriented methods, profit-oriented methods, and cost-oriented methods.

Household buildings are not considered in this indicator.

For smart water management at the household scale, see indicator 12.2.

A smart water meter should refer to a water meter that includes a real *-time* digital display or that is available through a real *-time online application*, so that customers can better understand their water usage.

Smart meters can digitally transmit their readings to water suppliers for more accurate water billing, and for water planning and conservation by the provider.

23.4.3 Data sources

24. Reporting and record keeping

Reports on city indicators should collect the necessary data in the individual test methods used.

APPENDIX A

(Information)

Mapping ISO 37122 indicators to issues and objectives in ISO 37101

Problems in ISO 37101	The purpose of this document
Governance, empowerment and engagement	Attraction (ISO 37101) — <u>5.1</u> Percentage of service contracts providing city services that contain a data transparency policy — <u>9.2</u> Percentage of payments to the city paid electronically based on electronic billing — <u>10.1</u> Annual number of visits to the <i>open-data</i> portal city government per 100,000 population — <u>10.2</u> Percentage of city services that can be accessed and obtained online — <u>10.3</u> Average response time for service requests made through the city's non-emergency service request system (days) — <u>13.2</u> Percentage of the city budget allocated to providing assistance, equipment, and technology to assist the mobility of city residents with special needs — <u>13.4</u> Percentage of city budget allocated to providing programs designed to bridge the digital <i>divide</i> — <u>21.2</u> Percentage of building permits registered through the electronic system — <u>21.3</u> Average time for building permit approval (days) Social cohesion (ISO 37101) — <u>10.2</u> Percentage of city services that can be accessed and obtained online — <u>13.2</u> Percentage of city budget allocated to providing assistance, equipment, and technology to assist the mobility of city residents with special needs — <u>13.4</u> Percentage of city budget allocated to providing programs designed to bridge the digital divide — <u>21.1</u> Number of city residents per 100,000 population per year involved in the planning process — Well-being (ISO 37101) — <u>10.3</u> Average response time to service requests (<i>inquiries</i>) made through the city's non-emergency service request system (days) — <u>13.2</u> Percentage of the city budget allocated to providing assistance, equipment, and technology to assist the mobility of city residents with special needs — <u>13.4</u> Percentage of city budget allocated to providing programs designed to bridge the digital <i>divide</i> Responsible use of resources (<i>Responsible resource use</i>) (ISO 37101)

Problems in ISO 37101	The purpose of this document
	— <u>5.1</u> Percentage of service contracts providing city services that contain a data transparency policy — <u>9.2</u> Percentage of payments to the city paid electronically based on electronic billing — <u>21.3</u> Average time for building permit approval (days) Resilience (ISO 37101) Environmental preservation and improvement (<i>Preservation and improvement of environment</i>)
Education and capacity building <i>Education and capacity building</i>	Attractiveness (ISO 37101) — <u>5.3</u> Percentage of the workforce employed in the information and communications technology (ICT) sector — <u>6.1</u> Percentage of city residents with professional fluency in more than one language — <u>6.2</u> Number of computers, laptops, tablets or other digital learning devices available per 1,000 students Social cohesion (ISO 37101) — <u>6.1</u> Percentage of city residents with professional fluency in more than one language — <u>17.1</u> Number of online bookings for cultural facilities per 100,000 inhabitants — <u>17.3</u> Number of book and e-book titles in public libraries per 100,000 population — <u>17.4</u> Percentage of city residents who are active library users Well-being (ISO 37101) — <u>6.1</u> Percentage of city residents with professional fluency in more than one language — <u>17.1</u> Number of online bookings for cultural facilities per 100,000 inhabitants — <u>17.3</u> Number of book and e-book titles in public libraries per 100,000 population — <u>17.4</u> Percentage of city residents who are active library users Responsible use of resources (<i>Responsible resource use</i>) (ISO 37101) Resilience (ISO 37101) — <u>5.3</u> Percentage of workforce employed in the information and communications technology (ICT) sector — <u>6.1</u> Percentage of city residents with professional fluency in more than one language — <u>6.3</u> Number of science, technology, engineering, and mathematics higher education degrees per 100,000 population Environmental preservation and improvement (<i>Preservation and improvement of environment</i>)

Problems in ISO 37101	The purpose of this document
Innovation, creativity and research	Attraction (ISO 37101) — <u>5.1</u> Percentage of service contracts providing city services that contain a data transparency policy — <u>5.2</u> New business continuity rate per 100,000 population — <u>5.3</u> Percentage of workforce employed in the information and communications technology (ICT) sector — <u>5.4</u> Percentage of the workforce employed in education, and research and development — 6.3 Number of higher education degrees in science, technology, engineering, and mathematics per 100,000 population Social cohesion (ISO 37101) Well-being (ISO 37101) Responsible use of resources (ISO 37101) — <u>5.1</u> Percentage of service contracts providing city services that contain a data transparency policy Durability (ISO 37101) — <u>5.2</u> New business continuity rate per 100,000 population — <u>5.3</u> Percentage of workforce employed in the information and communications technology (ICT) sector — <u>5.4</u> Percentage of the workforce employed in education, and research and development — 6.3 Number of science, technology, engineering and mathematics higher education degrees per 100,000 population Environmental preservation and improvement
Health and care in society	Attraction (ISO 37101) — <u>7.10</u> Number of electric vehicle charging stations per number of registered electric vehicles — <u>8.2</u> Number of real -time air quality monitoring stations per square kilometer (km²) — <u>19.3</u> Percentage of registered vehicles in the city that are low emission vehicles — <u>23.1</u> Percentage of drinking water under water quality monitoring through real-time water quality monitoring stations (<i>real-time</i>) — <u>23.2</u> Number of real -time environmental water quality monitoring stations per 100,000 population — <u>23.3</u> Percentage of city water distribution network monitored through smart water systems Social cohesion (ISO 37101) — <u>11.2</u> Annual number of remote medical registrations per 100,000 population

Problems in ISO 37101	The purpose of this document
	Well-being (ISO 37101) — <u>7.10</u> Number of electric vehicle charging stations per number of registered electric vehicles — <u>8.2</u> Number of real-time air quality monitoring stations per square kilometer (km²) — <u>8.3</u> Percentage of public buildings equipped with indoor air quality monitoring equipment — <u>11.1</u> Percentage of city residents who have integrated online health data that can be accessed by health service providers — <u>11.2</u> Annual number of remote medical registrations per 100,000 population — <u>11.3</u> Percentage of city residents with access to real-time public warning systems for air and water quality — <u>19.3</u> Percentage of registered vehicles in the city that are low emission vehicles Responsible use of resources (ISO 37101) — <u>23.1</u> Percentage of drinking water under water quality monitoring through real-time water quality monitoring stations — <u>23.3</u> Percentage of city water distribution network monitored through smart water systems Durability (ISO 37101) — <u>11.1</u> Percentage of city residents who have integrated online health data that can be accessed by health service providers — <u>11.2</u> Annual number of remote medical registrations per 100,000 population — <u>11.3</u> Percentage of city residents with access to real-time public warning systems for air and water quality — <u>20.3</u> Percentage of city areas covered by the online food supplier mapping system — <u>23.1</u> Percentage of drinking water under real-time water quality monitoring through water quality monitoring stations — <u>23.2</u> Number of real-time environmental water quality monitoring stations per 100,000 population — <u>23.3</u> Percentage of city water distribution network monitored through smart water systems Environmental preservation and improvement — <u>7.10</u> Number of electric vehicle charging stations per number of registered electric vehicles — <u>8.2</u> Number of real-time air quality monitoring stations per square kilometer (km²) — <u>19.3</u> Percentage of registered vehicles in the city that are low emission vehicles

Problems in ISO 37101	The purpose of this document
	— <u>20.1</u> Annual percentage of city budget allocated to urban agriculture initiatives — 23.2 Number of real <i>-time</i> environmental water quality monitoring stations per 100,000 residents
Culture and community identity	Attraction (ISO 37101) — <u>17.2</u> Percentage of the city's cultural records that have been digitized Social cohesion (ISO 37101) — <u>17.1</u> Number of online reservations for cultural facilities per 100,000 inhabitants <i>Well-being</i> (ISO 37101) — <u>17.1</u> Number of online reservations for cultural facilities per 100,000 inhabitants Responsible use of resources (ISO 37101) Durability (ISO 37101) — <u>17.1</u> Number of <i>online</i> reservations for cultural facilities per 100,000 inhabitants Environmental preservation and improvement
Coexistence, interdependence, and mutualism	Attraction (ISO 37101) — <u>13.1</u> Percentage of public buildings accessible to people with disabilities — <u>13.2</u> Percentage of the city budget allocated to providing assistance, equipment, and technology to assist the movement of city residents with special needs — 13.3 Percentage of pedestrian crossings equipped with easily accessible pedestrian markers — 13.4 Percentage of the city budget allocated to providing programs designed to bridge the digital <i>divide</i> — <u>19.2</u> Number of sharing economy transportation users <i>per</i> 100,000 population Social cohesion (ISO 37101) — <u>13.1</u> Percentage of public buildings accessible to people with disabilities — <u>13.2</u> Percentage of the city budget allocated to providing assistance, equipment, and technology to assist the movement of city residents with special needs — 13.3 Percentage of pedestrian crossing markings that have been equipped with easily accessible pedestrian markers

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Problems in ISO 37101	The purpose of this document
	— <u>13.4</u> Percentage of city budget allocated to providing programs designed to bridge the digital <i>divide</i> — <u>19.2</u> Number of sharing economy transportation users <i>per</i> 100,000 population Well-being (ISO 37101) — <u>13.1</u> Percentage of public buildings accessible to people with disabilities — <u>13.2</u> Percentage of the city budget allocated to providing assistance, equipment, and technology to assist the movement of city residents with special needs — 13.3 Percentage of pedestrian crossings equipped with easily accessible pedestrian markers — 13.4 Percentage of the city budget allocated to providing programs designed to bridge the digital <i>divide</i> — <u>19.2</u> Number of sharing economy transportation users <i>per</i> 100,000 population Responsible use of resources (ISO 37101) Toughness (ISO 37101) Environmental preservation and improvement (ISO 37101)
Sustainable economy and production and consumption	Attraction (ISO 37101) — <u>5.1</u> Percentage of service contracts providing city services that contain a data transparency policy — 5.2 New <u>business</u> start-up rate per 100,000 residents — <u>5.3</u> Percentage of workforce employed in the information and communications technology (ICT) sector — <u>5.4</u> Percentage of the workforce employed in education, and research and development — 7.8 Percentage of public buildings in need of renovation/repair — 7.9 Percentage of buildings in the city with smart energy meters — 9.1 Annual <u>amount</u> of income earned from the sharing economy as a share of own income sources — <u>9.2</u> Percentage of payments to the city paid electronically based on electronic billing — <u>12.1</u> Percentage of households with smart energy meters — <u>12.2</u> Percentage of households that have smart water meters — <u>21.3</u> Average time for building permit approval (days) — <u>23.1</u> Percentage of drinking water under water quality monitoring through real-time water quality monitoring stations (<i>real-time</i>)

Problems in ISO 37101	The purpose of this document
	— <u>23.3</u> Percentage of city water distribution network monitored through smart water system — <u>23.4</u> Percentage of buildings in the city that have smart water meters Social cohesion (ISO 37101) Well-being (ISO 37101) — <u>20.3</u> Percentage of city area covered by the online food supplier mapping system Responsible use of resources (ISO 37101) Percentage of service contracts providing city services that contain a data transparency policy — <u>7.1</u> Percentage of electrical and thermal energy produced from wastewater treatment, solid waste, and liquid waste treatment and other heat resources from waste, as a share of the city's total energy in annual units — 7.2 Electrical and thermal energy (GJ) produced from wastewater treatment per capita per year — <u>7.3</u> Electrical and thermal energy (GJ) produced from solid waste processing per capita per year — <u>7.4</u> Percentage of city electricity produced through decentralized electricity production systems — <u>7.5</u> Storage capacity of the city energy grid per total city energy consumption — <u>7.6</u> Percentage of street lighting managed through a light performance management system — <u>7.7</u> Percentage of street lighting that has been updated and newly installed — 7.8 Percentage of public buildings that require renovation/repair — 7.9 Percentage of buildings in the city that have smart energy meters — 9.2 Percentage of payments to the city that are paid electronically based on electronic bills — <u>12.1</u> Percentage of households that have smart energy meters — 12.2 Percentage of households that have smart water meters — <u>16.1</u> Percentage of waste collection centers equipped with telemetering — <u>16.2</u> Percentage of the city population that has door-to-door waste collection with individual monitoring of household waste quantities — <u>16.3</u> Percentage of the total amount of waste in the city that is used for energy generation — <u>16.4</u> Percentage of the total amount of plastic recycled in the city — <u>16.5</u> Percentage of scannable general waste bins — 16.6 Percentage of municipal electrical and electronic waste recycled

Problems in ISO 37101	The purpose of this document
	— <u>20.2</u> Total annual amount of municipal food waste sent to composting facilities per capita (in tonnes) — <u>21.3</u> Average time for building permit approval (days) — <u>22.1</u> Percentage of treated wastewater that is reused — <u>22.2</u> Percentage of <i>biosolids</i> reused (<i>dry matter mass</i>) — <u>22.3</u> Energy generated from wastewater as a percentage of total city energy consumption — <u>22.4</u> Percentage of the total amount of wastewater in the city used to generate energy — <u>22.5</u> Percentage of wastewater pipe network monitored by real-time tracking sensor systems — <u>23.1</u> Percentage of drinking water under water quality monitoring through real-time water quality monitoring stations (<i>real-time</i>) — <u>23.3</u> Percentage of city water distribution network monitored through smart water system — <u>23.4</u> Percentage of buildings in the city that have smart water meters (<i>smart water systems</i>) Toughness (ISO 37101) — <u>5.2</u> Survival rate of <i>new businesses</i> per 100,000 population — <u>5.3</u> Percentage of the workforce employed in the information and communications technology (ICT) sector — <u>5.4</u> Percentage of the workforce employed in the education, and research and development sectors — <u>7.4</u> Percentage of municipal electricity produced through decentralized electricity production systems — <u>7.5</u> Storage capacity of the city energy grid per total city energy consumption — <u>9.1</u> Annual amount of income earned from the sharing economy as part of own income sources — <u>16.3</u> Percentage of the total amount of waste in the city that is used for energy generation — <u>20.2</u> Total annual amount of municipal food waste sent to composting per capita (in tonnes) — <u>20.3</u> Percentage of city area covered by the online food supplier mapping system — <u>22.1</u> Percentage of treated wastewater that is reused — <u>22.2</u> Percentage of <i>biosolids</i> reused (<i>dry matter mass</i>) — <u>22.3</u> Energy generated from wastewater as a percentage of total city energy consumption — <u>22.4</u> Percentage of the total amount of wastewater in the city used to generate energy — <u>22.5</u> Percentage of wastewater pipeline network monitored through real-time tracking sensor systems (<i>real-time</i>)

Problems in ISO 37101	The purpose of this document
	— <u>23.1</u> Percentage of drinking water under water quality monitoring through real-time water quality monitoring stations (<i>real-time</i>) — <u>23.3</u> Percentage of city water distribution network monitored through smart water system Environmental preservation and improvement — <u>16.1</u> Percentage of waste collection centers equipped with <i>telemetering</i> — <u>16.2</u> Percentage of the city population that has door-to-door waste collection with individual monitoring of household waste quantities — <u>16.3</u> Percentage of the total amount of waste in the city that is used for energy generation — <u>16.4</u> Percentage of the total amount of plastic recycled within the city — <u>16.5</u> Percentage of <i>sensor-enabled</i> public waste bins — <u>16.6</u> Percentage of municipal electrical and electronic waste recycled — <u>20.1</u> Percentage of the city's annual budget allocated to urban agriculture <i>initiatives</i> — <u>22.1</u> Percentage of treated wastewater that is reused — <u>22.2</u> Percentage of <i>biosolids</i> reused (<i>dry matter mass</i>) — <u>22.3</u> Energy generated from wastewater as a percentage of total city energy consumption — <u>22.4</u> Percentage of the total amount of wastewater in the city used to generate energy — <u>22.5</u> Percentage of wastewater pipe network monitored by real -time <i>data-tracking sensor system</i>
Living and working environment	Attraction (ISO 37101) — 7.8 Percentage of public buildings requiring renovation/repair — 7.10 Number of electric vehicle charging stations per number of registered electric vehicles — 8.1 Percentage of buildings constructed or renovated in the last 5 years that adhere to green building principles — 8.2 Number of real-time air quality monitoring stations per square kilometer (km²) — 10.3 Average response time to service requests (<i>inquiries</i>) made through the city's non-emergency service request system (days) — 13.1 Percentage of public buildings accessible to people with disabilities — 19.3 Percentage of registered vehicles in the city that are low emission vehicles

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Problems in ISO 37101	The purpose of this document
	— 21.4 Percentage of urban population living in areas with medium to high population density — 23.2 Number of real <i>-time environmental water</i> quality monitoring stations per 100,000 population Social cohesion (ISO 37101) — 13.1 Percentage of public buildings accessible to people with disabilities — 21.4 Percentage of urban population living in areas with medium to high population density Well-being (ISO 37101) — 7.10 Number of electric vehicle charging stations per number of registered electric vehicles — 8.1 Percentage of buildings constructed or renovated in the last 5 years that adhere to green building principles — 8.2 Number of real-time air quality monitoring stations per square kilometer (km²) — 8.3 Percentage of public buildings equipped with indoor air quality monitoring equipment — 10.3 Average response time for service requests made through the city's non-emergency service request system (days) — 13.1 Percentage of public buildings accessible to people with disabilities — 14.1 Percentage of public recreation services that can be booked <i>online</i> — 19.3 Percentage of registered vehicles in the city that are low emission vehicles — 21.4 Percentage of urban population living in areas with medium to high population density Responsible use of resources (ISO 37101) — 7.8 Percentage of public buildings requiring renovation/repair — 8.1 Percentage of buildings constructed or renovated in the last 5 years that adhere to <i>green</i> building principles Resilience (ISO 37101) — 23.2 Number of real <i>-time environmental water</i> quality monitoring stations per 100,000 population Environmental preservation and improvement (ISO 37101) — 7.10 Number of electric vehicle charging stations per number of registered electric vehicles — 8.2 Number of real <i>-time</i> air quality monitoring stations per square kilometer (km²)

Problems in ISO 37101	The purpose of this document
	— 19.3 Percentage of registered vehicles in the city that are low emission vehicles — 23.2 Number of real <i>-time environmental water</i> quality monitoring stations per 100,000 population
Safety and security	Attractiveness (ISO 37101) — 9.2 Percentage of payments to the city paid electronically based on electronic billing — 10.4 Average downtime of city IT infrastructure — 15.1 Percentage of city area covered by digital surveillance cameras Social cohesion (ISO 37101) Well-being (ISO 37101) — 15.1 Percentage of city area covered by digital surveillance cameras Responsible use of resources (ISO 37101) — 7.6 Percentage of street lighting managed through a light performance management system — 9.2 Percentage of payments to the city paid electronically based on electronic billing Durability (ISO 37101) — 10.4 Average downtime of city IT infrastructure Environmental preservation and improvement
Community infrastructure	Attractions (ISO 37101) — 7.9 Percentage of buildings in the city that have smart energy meters — 8.1 Percentage of buildings built or renovated in the last 5 years that adhere to green building principles — 10.4 Average <i>downtime</i> average city IT infrastructure — 12.1 Percentage of households that have smart energy meters — 12.2 Percentage of households that have smart water meters — 13.3 Percentage of pedestrian crossing markings that are equipped with accessible pedestrian <i>signals</i> — 18.1 Percentage of city residents who have access to fast enough <i>broadband</i> internet — 18.2 Percentage of city area that does not have a telecommunications network

Problems in ISO 37101	The purpose of this document
	— 18.3 Percentage of city area covered by internet access provided by the city government — 19.1 Percentage of city streets and highways covered by <i>real-time</i> online traffic information and warnings — 19.4 Number of bicycles available through bicycle rental services organized by the city government per 100,000 inhabitants — 19.5 Percentage of public transport lines equipped with publicly accessible real-time systems — 19.6 Percentage of the city's public transport network covered by an integrated payment system — 19.7 Percentage of public parking spaces equipped with an electronic payment system — 19.9 Percentage of smart traffic lights — 19.12 Percentage of public transportation routes with internet connectivity for users of such transportation, which are provided and/or managed by the city government — 19.13 Percentage of roads compatible with autonomous driving systems — 19.14 Percentage of city bus fleets using electric drives — 23.1 Percentage of drinking water under water quality monitoring through real-time water quality monitoring stations — 23.2 Number of real-time environmental water quality monitoring stations per 100,000 population — 23.3 Percentage of city water distribution network monitored through smart water <i>system</i> — 23.4 Percentage of buildings in the city that have smart water meters Social cohesion (ISO 37101) — 13.3 Percentage of pedestrian crossings that are equipped with easily accessible pedestrian markers— 18.1 Percentage of city residents who have access to fast enough <i>broadband</i> internet — 18.2 Percentage of city area that does not have a telecommunications network — 18.3 Percentage of city area covered by internet access provided by the city government — 19.4 Number of bicycles available through bicycle rental services organized by the city government per 100,000 inhabitants — 19.5 Percentage of public transport lines equipped with accessible <i>real-time</i> systems in general — 19.6 Percentage of the city's public transport network covered by an integrated payment system — 19.9 Percentage of smart traffic lights

Problems in ISO 37101	The purpose of this document
	Well-being (ISO 37101) — 8.1 Percentage of buildings constructed or renovated in the last 5 years that adhere to green building principles — 13.3 Percentage of pedestrian crossings equipped with easily accessible pedestrian signs — 19.1 Percentage of city streets and highways covered by real-time online traffic information and <i>alerts</i> — 19.4 Number of bicycles available through bicycle rental services organized by the city government per 100,000 inhabitants — 19.5 Percentage of public transport lines equipped with publicly accessible real-time systems — 19.6 Percentage of the city's public transport network covered by a unified payment <i>system</i> — 19.7 Percentage of public parking spaces equipped with electronic payment systems — 19.8 Percentage of public parking spaces equipped with real-time availability information systems — 19.9 Percentage of smart traffic lights — 19.10 Percentage of city area mapped with real-time interactive map system — 19.12 Percentage of public transportation routes with internet connectivity for users of such transportation, which are provided and/or managed by the city government Responsible use of resources (ISO 37101) — 7.1 Percentage of electrical and thermal energy produced from wastewater treatment, solid waste, and liquid waste treatment and other heat resources from waste, as a share of the city's total energy in annual units — 7.2 Electrical and thermal energy (GJ) produced from wastewater treatment per capita per year — 7.3 Electrical and thermal energy (GJ) produced from solid waste processing per capita per year — 7.4 Percentage of city electricity produced through decentralized electricity production systems — 7.5 Storage capacity of the city energy grid per total city energy consumption — 7.9 Percentage of buildings in the city that have smart energy meters — 8.1 Percentage of buildings built or renovated in the last 5 years that adhere to green building principles — 12.1 Percentage of households that have smart energy meters

Problems in ISO 37101	The purpose of this document
	— 12.2 Percentage of households that have smart water meters — 16.1 Percentage of waste collection centers equipped with telemetering — 16.2 Percentage of the city's population that has door-to-door waste collection with individual monitoring of the amount of household waste — 16.3 Percentage of the total amount of waste in the city that is used for energy generation — 16.4 Percentage of the total amount of plastic recycled in the city — 16.5 Percentage of scannable general waste bins — 16.6 Percentage of municipal electrical and electronic waste recycled — 22.1 Percentage of treated wastewater used return — 22.2 Percentage of biosolids reused (<i>dry matter mass</i>) — 22.3 Energy generated from wastewater as a percentage of total city energy consumption — 22.4 Percentage of the total amount of wastewater in the city used to generate energy — 22.5 Percentage of wastewater pipe network monitored through real-time data tracking sensor system (<i>real-time</i>) — 23.1 Percentage of drinking water under water quality monitoring through real-time water quality monitoring stations (<i>real-time</i>) — 23.3 Percentage of city water distribution networks monitored through smart water systems — 23.4 Percentage of buildings in the city that have smart water meters Toughness (ISO 37101) — 7.4 Percentage of city electricity produced through decentralized electricity production systems — 7.5 Storage capacity of the city energy grid per total city energy consumption — 10.4 Average downtime of city IT infrastructure — 16.3 Percentage of the total amount of waste in the city that is used for energy generation — 22.1 Percentage of treated wastewater that is reused — 22.2 Percentage of <i>biosolids</i> reused (<i>dry matter mass</i>) — 22.3 Energy generated from wastewater as a percentage of total city energy consumption — 22.4 Percentage of the total amount of wastewater in the city used to generate energy — 22.5 Percentage of wastewater pipe network monitored through real-time data tracking sensor systems

Problems in ISO 37101	The purpose of this document
	— 23.1 Percentage of drinking water under water quality monitoring through real-time water quality monitoring stations (<i>real-time water quality monitoring station</i>) — 23.2 Number of real -time <i>environmental water</i> quality monitoring stations per 100,000 population — 23.3 Percentage of city water distribution network monitored through smart water system Environmental preservation and improvement — 16.1 Percentage of waste collection centers equipped with telemetering — 16.2 Percentage of the city's population that has door-to-door waste collection with individual monitoring of the amount of household waste — 16.3 Percentage of the total amount of waste in the city that is used for energy generation — 16.4 Percentage of the total amount of plastic recycled within the city — 16.5 Percentage of scannable general waste bins — 16.6 Percentage of municipal electrical and electronic waste recycled — 19.1 Percentage of city streets and highways covered by live online traffic alerts and information — 19.8 Percentage of public parking spaces equipped with real-time availability information systems — 19.10 Percentage of city area mapped through a real-time interactive map system — 22.1 Percentage of treated wastewater that is reused — 22.2 Percentage of <i>biosolids</i> reused (<i>dry matter mass</i>) — 22.3 Energy generated from wastewater as a percentage of total city energy consumption — 22.4 Percentage of the total amount of wastewater in the city used to generate energy — 22.5 Percentage of wastewater pipe network monitored by real-time data tracking sensor systems — 23.2 Number of real-time environmental water quality monitoring stations <i>per</i> 100,000 population
Mobility Mobility	Attraction (ISO 37101) — 19.1 Percentage of city streets and highways covered by live online traffic alerts and information — 19.2 Number of sharing economy transportation users per 100,000 population — 19.4 Number of bicycles available through bicycle rental services organized by the city government per 100,000 inhabitants

Problems in ISO 37101	The purpose of this document
	— 19.5 Percentage of public transport lines equipped with a live time system that is accessible online general — 19.6 Percentage of the city's public transport network covered by an integrated payment system — 19.7 Percentage of public parking spaces equipped with an electronic payment system — 19.12 Percentage of public transportation routes with internet connectivity for users of such transportation, which are provided and/or managed by the city government — 19.13 Percentage of the number of roads that comply with the system autonomous driving — 19.14 Percentage of city bus fleet using electric drive Social cohesion (ISO 37101) — 19.2 Number of sharing economy transportation users per 100,000 population — 19.4 Number of bicycles available through bicycle rental services organized by the city government per 100,000 inhabitants — 19.5 Percentage of public transport lines equipped with publicly accessible real-time systems — 19.6 Percentage of the city's public transport network covered by an integrated payment system Well-being (ISO 37101) — 19.1 Percentage of city streets and highways covered by <i>real-time</i> online traffic alerts and information — 19.2 Number of <i>sharing economy</i> transportation users per 100,000 population — 19.4 Number of bicycles available through bicycle rental services organized by the city government per 100,000 inhabitants — 19.5 Percentage of public transport lines equipped with publicly accessible real-time systems — 19.6 Percentage of the city's public transport network covered by an integrated payment system — 19.7 Percentage of public parking spaces equipped with an electronic payment system — 19.8 Percentage of public parking spaces equipped with real-time availability information systems — 19.10 Percentage of city area mapped with real-time interactive map system — 19.12 Percentage of public transportation routes with internet connectivity for users of such transportation, which are provided and/or managed by the city government Responsible use of resources (ISO 37101)

Problems in ISO 37101	The purpose of this document
	— 16.1 Percentage of waste collection centers equipped with <i>telemetering</i> — 16.2 Percentage of the city population that has door-to-door waste collection with individual monitoring of the amount of household waste Resilience (ISO 37101) Environmental preservation and improvement — 16.1 Percentage of waste collection centers equipped with <i>telemetering</i> — 16.2 Percentage of city population that has door-to-door waste collection with individual monitoring of domestic waste quantity — 19.1 Percentage of city streets and highways covered by <i>real-time</i> online traffic alerts and information — 19.8 Percentage of public parking spaces equipped with real-time availability information systems — 19.10 Percentage of city area mapped with live interactive map system — 19.11 Percentage of registered vehicles in the city that are autonomous vehicles
Biodiversity and ecosystem services Biodiversity and ecosystem services	Attraction (ISO 37101) — 23.2 Number of <i>real-time environmental water</i> quality monitoring stations per 100,000 population Social cohesion (ISO 37101) Well-being (ISO 37101) Responsible use of resources (ISO 37101) — 22.1 Percentage of treated wastewater that is reused — 22.2 Percentage of <i>biosolids</i> reused (<i>dry matter mass</i>) Toughness (ISO 37101) — 22.1 Percentage of treated wastewater that is reused — 22.2 Percentage of <i>biosolids</i> reused (<i>dry matter mass</i>) — 23.2 Number of real-time environmental water quality monitoring stations per 100,000 population Environmental preservation and improvement — 22.1 Percentage of treated wastewater that is reused — 22.2 Percentage of <i>biosolids</i> reused (<i>dry matter mass</i>)

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Problems in ISO 37101	The purpose of this document
	— 23.2 Number of real-time environmental water quality monitoring stations per 100,000 population

APPENDIX B

(Information)

Mapping indicators against *the Sustainable Development Goals (SDGs)* United Nations (2015)

Sustainable Development Goals (2015)	Indicators
Goal 1: End poverty in all its forms worldwide	— <u>13.2</u> Percentage of the city budget allocated to providing assistance, equipment, and technology to assist the mobility of city residents with special needs — <u>13.4</u> Percentage of city budget allocated to providing programs designed to bridge the digital <i>divide</i>
Goal 2: End hunger, achieve food security and improved nutrition and promote sustainable agriculture	— <u>20.1</u> Annual percentage of city budget allocated to urban agriculture initiatives — 20.2 Total amount of city food waste collected to composting facilities per capita (in tons) — <u>20.3</u> Percentage of city area covered by the online food supplier mapping system
Goal 3: Ensure healthy lives and promote well-being for all at all ages	— <u>8.2</u> Number of real -time remote air quality monitoring stations per square kilometer (km²) — <u>8.3</u> Percentage of public buildings equipped with indoor air quality monitoring equipment — <u>11.1</u> Percentage of city residents who have integrated online health data that can be accessed by health service providers — <u>11.2</u> Annual number of remotely conducted medical registrations per 100,000 population — <u>11.3</u> Percentage of city residents with access to real-time public warning systems for air and water quality — <u>23.1</u> Percentage of drinking water under water quality monitoring through real-time water quality monitoring stations — 23.2 Number of real-time environmental water quality monitoring stations per 100,000 population — <u>23.3</u> Percentage of city water distribution network monitored through smart water systems
Goal 4: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all	— <u>6.1</u> Percentage of city residents with professional fluency in more than one language — <u>6.2</u> Number of computers, laptops, tablets or other digital learning devices available per 1,000 students — <u>6.3</u> Number of science , <i>technology, engineering, and mathematics</i> (STEM) higher education degrees per 100,000 population

Sustainable Development Goals (2015)	Indicators
	— <u>13.4</u> Percentage of city budget allocated to providing programs designed to bridge the digital <i>divide</i> — <u>17.3</u> Number of book and e-book titles in public libraries per 100,000 population — <u>17.4</u> Percentage of city residents who are active library users
Goal 5: Achieve gender equality and empower all women and girls	-
Goal 6: Ensure availability and sustainable management of water and sanitation for all everyone	— <u>12.2</u> Percentage of households that have smart water meters — <u>22.1</u> Percentage of treated wastewater that is reused — <u>22.2</u> Percentage of <i>biosolids</i> reused (<i>dry matter mass</i>) — <u>22.5</u> Percentage of wastewater pipe network monitored through real-time data tracking sensor systems — <u>23.1</u> Percentage of drinking water under water quality monitoring through real-time water quality monitoring stations (<i>real-time water quality monitoring station</i>) — <u>23.2</u> Number of real <i>-time environmental water</i> quality monitoring stations per 100,000 population — <u>23.3</u> Percentage of city water distribution networks monitored through smart water <i>systems</i> — <u>23.4</u> Percentage of buildings in the city that have smart water meters
Goal 7: Ensure access to affordable, reliable, sustainable and modern energy for all	— <u>7.1</u> Percentage of electrical and thermal energy produced from wastewater treatment, solid waste, and liquid waste treatment and other heat resources from waste, as a share of the city's total energy in annual units — <u>7.2</u> Electrical and thermal energy (GJ) produced from wastewater treatment per capita per year — <u>7.3</u> Electrical and thermal energy (GJ) produced from solid waste processing per capita per year — <u>7.4</u> Percentage of city electricity produced through decentralized electricity production systems — <u>7.5</u> Storage capacity of the city energy grid per total city energy consumption — <u>7.6</u> Percentage of street lighting managed through a light performance management system — <u>7.7</u> Percentage of street lighting that has been updated and newly installed

Sustainable Development Goals (2015)	Indicators
	— <u>7.8</u> Percentage of public buildings requiring renovation/repair — 7.9 Percentage of <u>buildings</u> in the city that have smart energy meters — <u>8.1</u> Percentage of buildings constructed or renovated in the last 5 years that adhere to green building principles — 12.1 Percentage of households that have smart energy meters — — <u>22.3</u> Energy generated from wastewater as a percentage of total city energy consumption — <u>22.4</u> Percentage of the total amount of wastewater in the city used to generate energy — <u>16.3</u> Percentage of the total amount of waste in the city that is used for energy generation
Goal 8: Promote strong, inclusive and sustainable economic growth and productive and decent work for all	— <u>5.2</u> New business continuity rate per 100,000 population — <u>5.3</u> Percentage of the workforce employed in the information and communications technology (ICT) sector — <u>5.4</u> Percentage of the workforce employed in the education, and research and development sectors — 6.1 Percentage of the city's <u>s</u> population with professional fluency in more than one language — <u>6.3</u> Number of science , <i>technology, engineering</i>, and mathematics (STEM) higher education degrees per 100,000 population — <u>13.4</u> Percentage of city budget allocated to providing programs designed to bridge the digital <i>divide</i>
Goal 9: Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation	— <u>5.3</u> Percentage of workforce employed in the information and communication technology (ICT) sector — <u>5.4</u> Percentage of the workforce employed in the education, and research and development sectors — 6.3 Number of science, <i>technology, engineering</i>, and mathematics (<i>STEM</i>) higher education degrees / STEM) per 100,000 population — <u>7.5</u> Storage capacity of the city energy grid per total city energy consumption — <u>10.4</u> Average downtime of city IT infrastructure — <u>13.4</u> Percentage of city budget allocated to providing programs designed to bridge the digital <i>divide</i> — <u>18.1</u> Percentage of city residents who have access to fast enough <i>broadband</i> internet — <u>18.2</u> Percentage of city area that does not have a telecommunications network

Sustainable Development Goals (2015)	Indicators
	— <u>18.3</u> Percentage of city area covered by internet connectivity provided by the city government — <u>19.1</u> Percentage of city streets and highways covered by <i>real-time</i> online traffic information and warnings — <u>19.5</u> Percentage of public transport lines equipped with publicly accessible real-time systems — <u>19.8</u> Percentage of public parking spaces equipped with real-time availability information systems — <u>19.9</u> Percentage of smart traffic lights — <u>19.12</u> Percentage of public transportation routes with internet connectivity for users of such transportation, which are provided and/or managed by the city government — <u>19.13</u> Percentage of roads suitable for autonomous driving systems — <u>22.5</u> Percentage of wastewater pipe networks monitored by real-time data tracking sensor systems — <u>23.1</u> Percentage of drinking water under water quality monitoring through real-time water quality monitoring stations — <u>23.2</u> Number of real-time environmental water quality monitoring stations per 100,000 inhabitants — <u>23.3</u> Percentage of city water distribution network monitored through smart water systems
Goal 10: Reduce inequality within and among countries	— <u>13.2</u> Percentage of the city budget allocated to providing assistance, equipment, and technology to assist the mobility of city residents with special needs — <u>13.4</u> Percentage of city budget allocated to providing programs designed to bridge the digital <i>divide</i>
Goal 11: Make cities and human settlements inclusive, safe, resilient and sustainable	— <u>7.10</u> Number of electric vehicle charging stations per number of registered electric vehicles — <u>8.1</u> Percentage of buildings constructed or renovated in the last 5 years that adhere to green building principles — <u>8.2</u> Number of real-time air quality monitoring stations per square kilometer (km²) — <u>8.3</u> Percentage of public buildings equipped with indoor air quality monitoring equipment — <u>10.2</u> Percentage of city services that can be accessed and obtained online — <u>11.3</u> Percentage of city residents with access to real-time public warning systems for air and water quality — <u>12.1</u> Percentage of households that have smart energy meters — <u>12.2</u> Percentage of households that have smart water meters

Sustainable Development Goals (2015)	Indicators
	— <u>13.1</u> Percentage of public buildings accessible to people with disabilities — <u>13.2</u> Percentage of the city budget allocated to providing assistance, equipment, and technology to assist the movement of city residents with special needs — 13.3 Percentage of pedestrian crossings equipped with accessible pedestrian markers easily — 14.1 Percentage of public recreation services that can be booked <i>online</i> — <u>15.1</u> Percentage of city area covered by digital surveillance cameras — <u>16.1</u> Percentage of waste collection centers equipped with <i>telemetering</i> — <u>16.2</u> Percentage of the city population that has door-to-door waste collection with individual monitoring of household waste quantities — <u>16.3</u> Percentage of the total amount of waste in the city that is used for energy generation — <u>16.4</u> Percentage of the total amount of plastic recycled within the city — <u>16.5</u> Percentage of <i>sensor-enabled</i> public waste bins — <u>16.6</u> Percentage of municipal electrical and electronic waste recycled — <u>17.1</u> Number of online reservations for cultural facilities per 100,000 inhabitants — <u>17.2</u> Percentage of the city's cultural records that have been digitized — <u>17.3</u> Number of book and e-book titles in public libraries per 100,000 population — <u>17.4</u> Percentage of city residents who are active library users — <u>19.1</u> Percentage of city streets and highways covered by <i>real-time</i> online traffic information and warnings — <u>19.3</u> Percentage of registered vehicles in the city that are low emission vehicles — <u>19.4</u> Number of bicycles available through bicycle rental services organized by the city government per 100,000 inhabitants — <u>19.5</u> Percentage of public transport lines equipped with a live time system that is accessible online general — <u>19.6</u> Percentage of the city's public transport network covered by an integrated payment system — <u>19.9</u> Percentage of smart traffic lights

Sustainable Development Goals (2015)	Indicators
	— <u>19.10</u> Percentage of city area mapped with live interactive map system — <u>19.13</u> Percentage of roads compatible with autonomous driving systems — 19.14 Percentage of the city bus fleet that uses electric propulsion — 20.1 Percentage of the city's annual budget allocated to urban agriculture initiatives — <u>21.1</u> Number of city residents per 100,000 inhabitants per year involved in the planning process — <u>21.4</u> Percentage of urban population living in areas with medium to high population density
Goal 12: Ensure sustainable consumption and production patterns	— <u>7.1</u> Percentage of electrical and thermal energy produced from wastewater treatment, solid waste, and liquid waste treatment and other heat resources from waste, as a share of the city's total energy in annual units — <u>7.2</u> Electrical and thermal energy (GJ) produced from wastewater treatment per capita per year — <u>7.3</u> Electrical and thermal energy (GJ) produced from solid waste processing per capita per year — <u>7.9</u> Percentage of buildings in the city that have smart energy meters — <u>7.10</u> Number of electric vehicle charging stations per number of registered electric vehicles — <u>16.1</u> Percentage of waste collection centers equipped with telemetering — <u>16.2</u> Percentage of the city population that has door-to-door waste collection with individual monitoring of household waste quantities — <u>16.3</u> Percentage of the total amount of waste in the city that is used for energy generation — <u>16.4</u> Percentage of the total amount of plastic recycled within the city — <u>16.5</u> Percentage of scannable public waste bins — <u>16.6</u> Percentage of municipal electrical and electronic waste recycled — <u>19.3</u> Percentage of registered vehicles in the city that are low emission vehicles — <u>20.1</u> Percentage of the city's annual budget allocated to urban agriculture initiatives — <u>20.2</u> Total amount of municipal food waste collected to composting facilities per capita (in tonnes) — <u>22.1</u> Percentage of treated wastewater that is reused — <u>22.2</u> Percentage of biosolids reused (dry matter mass) — <u>22.3</u> Energy generated from wastewater as a percentage of total city energy consumption

Sustainable Development Goals (2015)	Indicators
	— <u>22.4</u> Percentage of the total amount of wastewater in the city used to generate energy
Goal 13: Take urgent action to reduce climate change and its impacts	— <u>7.4</u> Percentage of city electricity produced through decentralized electricity production systems — <u>16.3</u> Percentage of the total amount of waste in the city that is used for energy generation — <u>22.1</u> Percentage of treated wastewater that is reused — <u>22.2</u> Percentage of <i>biosolids</i> reused (<i>dry matter mass</i>) — <u>22.3</u> Energy generated from wastewater as a percentage of total city energy consumption — <u>22.4</u> Percentage of the total amount of wastewater in the city used to generate energy — <u>23.3</u> Percentage of city water distribution networks monitored through smart water systems
Goal 14: Conserve and sustainably use marine resources for sustainable development	— <u>16.4</u> Percentage of the total amount of plastic recycled within the city — <u>22.5</u> Percentage of wastewater pipe network monitored through real-time data tracking sensor systems — 23.2 Number of real-time environmental water <u>quality</u> monitoring stations per 100,000 population
Goal 15: Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss.	— <u>21.4</u> Percentage of urban population living in areas with medium to high population density
Goal 16: Promote peaceful and inclusive societies for sustainable development, provide access to justice for all	— <u>5.1</u> Percentage of service contracts providing city services that contain a data transparency policy — <u>9.1</u> Annual amount of income earned from the sharing economy as a share of <i>own-source revenue</i>

Sustainable Development Goals (2015)	Indicators
everyone and establish effective, accountable and inclusive institutions at all levels	— <u>9.2</u> Percentage of payments to the city paid electronically based on electronic billing — <u>10.1</u> Annual number of online visits to the city's open data portal per 100,000 inhabitants — <u>10.2</u> Percentage of city services that can be accessed and obtained online — <u>10.3</u> Average response time for service requests made through the city's non-emergency service request system (days) — <u>21.1</u> Number of city residents per 100,000 inhabitants per year involved in the planning process — <u>21.2</u> Percentage of building permits collected through electronic systems — <u>21.3</u> Average time for building permit approval (days)
Goal 17: Strengthen the means of implementation and revitalize the global partnership for sustainable development	-

Supporting Information Regarding Standard Formulation

[1] SNI Technical Committee for Formulation

Technical Committee 13-11, Sustainable Cities and Communities

[2] Membership composition of the SNI Formulation Sub-

Chairman	Committee : Gensly,
Secretary	SE, MPA: Dina Nur Febriani, ST, MT
Member	: Dwi Elfrida Martina Simanungkalit, SIP, M.PPPA
	Syarif Hidayatullah, ST, MT
	Farid Subkhan, SAB, ME, M.Dev
	Hendra Sumiarsa, SE, MM
	Wahyudi, ST
	Billy Alkadia, SE
	Bambang Pramusinto, SH, S.IP, M.Si
	Eka Suryo Prihantoro, S.Si, M.Kom.
	Dana Indra Sense, Ph. D Dedy
	Permadi, SIP, MA, Ph.D
	Teddy Sukardi

[3] SNI design conceptor Sherly Dewi

Maharani, S.Si Wira Prasetya, ST,
MT

[4] Secretariat of the Komtek management for formulating SNI

Directorate of Infrastructure Standards Development, Conformity Assessment, Personnel, and
Creative Economy
National Standardization Agency